

Scale, Renormalization Group, and Emergent Symmetry

Quotient transport, fixed-point structure, effective envelopes,
and a realized vacuum-sector correspondence

SYM-10 of the Standard Model Symmetry Arc

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Principal result

Fix a renormalized QFT or EFT target

$$D_{10} = (\mathcal{T}_R, O_R, Q_{\text{phys}}, I_\mu, \mathfrak{S}, \mathfrak{E}, \mathfrak{M}, \mathfrak{B}, \mathfrak{B}, \mathfrak{G}, \mathfrak{A}),$$

where the entries specify the theory and operator basis, physical quotient, scale interval, lawful presentation class, effective envelope, threshold and matching data, background and boundary structure, state or phase, symmetry/action data, and the anomaly, Ward, fixed-point, conformal, emergent-symmetry, or vacuum-sector premise bundle. The manuscript proves that a physical dilatation, RG transport on theory space, a scheme or field-coordinate change, an EFT threshold, and a change of physical state are different typed operations. Scale-indexed descriptions descend through a presentation quotient to composable RG transport maps; fixed points and universality are quotient-stable structures; threshold matching glues distinct effective envelopes; dimensional transmutation produces an RG-invariant relation rather than an internally selected numerical scale; and an IR symmetry is a physical/effective action on a declared IR target, while a claim that it emerges from a specified UV theory carries the additional burden of a UV-to-IR transport and matching record. An arbitrary renormalization scale, cutoff, subtraction point, fitted threshold, naturalness criterion, bare coordinate, or unanchored vev cannot become a physical scale anchor without an independent witness. In the restricted homogeneous metric-proportional vacuum sector V_0 , finite renormalization shifts the matter-sector coefficient by an arbitrary constant. The manuscript proves that scalar AASC standing is equivalent there to descent to the ordinary QFT finite-renormalization quotient. Relational differences, response quantities, and the gravitationally renormalized total coefficient survive in distinct roles; the absolute matter-sector offset does not become a standalone prediction.

Document status	Publication edition v1.1.1
Proof order	Mainstream QFT statement; physical quotient or observable certificate; AASC role assignment; scope-preserving rival analysis
Direct SYM imports	SYM-0, SYM-2, SYM-3, SYM-5, SYM-6, SYM-7, and the global-target discipline exported by SYM-9
Realized-QFT bridge	Homogeneous vacuum finite-renormalization quotient and standing correspondence, restricted to V_0
Successor boundary	Scale-indexed atlas and role package to SYM-11; candidate completeness and whole-arc exhaustion remain downstream

Abstract

Scale transformations, renormalization-group flow, scheme changes, fixed-point symmetry, effective-field-theory matching, and emergent symmetry are frequently discussed with one vocabulary even though they act on different objects. This paper fixes a renormalized-QFT/EFT target carrying an operator basis, physical quotient, scale interval, presentation class, effective envelope, matching data, background, state or phase, and the premise bundle required for fixed-point or emergent-symmetry claims. It separates active dilatations on fields and states from RG transport in coupling/operator space and from scheme, regulator, normalization, and field-coordinate redescription.

The manuscript derives a renormalized trace identity that keeps explicit mass deformations, beta-function terms, anomalous dimensions, operator mixing, improvement, equations-of-motion/contact terms, and background contributions separately typed. It distinguishes the raw beta vector on coupling coordinates from the beta vector induced on the nonredundant physical quotient, so recurrent flavor or redundant motion is not mistaken for physical running. It then defines scale-indexed descriptions and proves that, under explicit overlap, quotient, continuation, envelope, and matching hypotheses, RG maps descend to composable maps on physical quotient data. Scheme changes are lawful presentation changes only when couplings, beta functions, operator bases, anomalous dimensions, Wilson coefficients, matching maps, and observable reconstruction are transported coherently.

Universality classes and fixed points are treated as quotient-stable structures rather than numerical coincidences among microscopic coordinates. The nonredundant spectrum of the stability matrix determines relevant, irrelevant, and marginal response classes; conformal enhancement remains conditional on the dimension, locality, unitarity, stress-tensor, spectrum, and virial-current assumptions actually used. Effective theories above and below a threshold are different envelopes connected by a matching morphism, while dimensional transmutation converts a running coordinate plus boundary or integration data into an RG-invariant scale relation without numerically selecting that scale. An IR symmetry is classified from its action, operator locus, scale window, state or phase, controlled breaking, and anomaly data. When the stronger claim is that this symmetry emerges from a specified UV theory, a separate ancestry record supplies the UV-to-IR matching map, deformation history, and anomaly transport.

A scale/no-promotion theorem blocks the use of the renormalization scale, a cutoff, a subtraction point, a fitted matching threshold, a bare parameter, a naturalness preference, or an unanchored vev as an independent physical scale selector. Finally, in the restricted homogeneous metric-proportional vacuum sector, the manuscript proves that AASC standing and descent to the standard finite-renormalization quotient are equivalent for scalar predictions. The absolute matter-sector vacuum coefficient is presentation-dependent; relational differences, response quantities, and the gravitationally renormalized total coefficient survive in distinct physical roles. The integrated result is a renormalization-covariant scale and emergent-symmetry normal form that distinguishes intrinsic IR symmetry from UV-emergence provenance and supplies a typed scale-sector handoff to SYM-11.

Keywords: scale symmetry; renormalization group; Callan–Symanzik equation; trace anomaly; scheme dependence; universality; fixed point; effective field theory; dimensional transmutation; emergent symmetry; vacuum energy; AASC.

Proof and certification status

The scientific conclusions are carried by the mathematical and physical argument in this manuscript and by explicitly identified primary-QFT and AASC imports. The accompanying project audits target scope, source authority, theorem identity, encoded algebra, table synchronization, bounded rival records, artifact identity, and reproducibility. It does not derive beta functions, formalize interacting QFT, prove candidate completeness, or replace the scientific proof with a validator.

Reader's guide: theorem spine, claim hierarchy, and exact endpoint

SYM-10 is the scale and renormalization stage of the Standard Model Symmetry Arc. Earlier modules fixed the quotient-action calculus, the active/passive spacetime distinction, anomaly typing, basis and spurion covariance, effective and emergent symmetry interfaces, operator-dimension scope, and global-form target discipline. The present manuscript uses those interfaces but does not inherit a ready-made trace identity, beta function, fixed-point theorem, or EFT matching calculation from them. Those physical results are established in mainstream QFT language before the AASC role map is applied.

The governing distinction is

active dilatation on fields or states $\neq$ RG transport on theory space
 $\neq$ scheme or coordinate redescription

The objects can be related. A Callan–Symanzik equation connects explicit scale differentiation to beta functions and anomalous dimensions; a fixed point may support a physical dilatation action; a scheme change pushes forward the RG vector field. But no one of these statements identifies the three loci.

Scientific proof spine

1. Freeze the renormalized theory, composite-operator basis, physical quotient, scale interval, scheme class, EFT envelope, threshold/matching data, background, state or phase, and the premise bundle for fixed-point or emergence claims.
2. Prove that constant-like quantities and parameter motions separate into finite role and mobility classes; running is lawful transport, not unrestricted numerical reassignment.
3. Derive the classical dilatation current and the conditions under which an improved stress tensor has vanishing trace.
4. Establish the renormalized trace identity with explicit, anomalous, mixing, improvement, EOM/contact, and background terms kept distinct.
5. Define scale-indexed descriptions and prove descent of RG transport to physical quotient data with identity, composition, and fail-closed boundary conditions.
6. Prove coherent scheme covariance for couplings, beta functions, operators, anomalous dimensions, Wilson coefficients, matching data, and observables.
7. Define universality quotients, quotient-stable fixed points by the induced nonredundant RG vector, and nonredundant relevant, irrelevant, and marginal response classes; keep raw flavor/redundant beta motion separate and conformal enhancement conditional.
8. Treat UV and IR EFTs as different envelopes glued by matching, then distinguish dimensional transmutation from absolute numerical scale selection.
9. Type an IR symmetry by its own IR target, scale window, operator locus, state or phase, controlled breaking, and anomaly data; require UV-to-IR matching and deformation ancestry only for the stronger claim that the symmetry emerges from a specified UV theory.
10. Prove the scale/no-promotion theorem.
11. In the restricted homogeneous vacuum sector, prove the finite-renormalization quotient theorem, the semiclassical split non-identifiability theorem, and the standing/quotient correspondence.
12. Apply the AASC role map and assemble the scale-sector normal form and SYM-11 handoff.

Claim hierarchy

The paper separates four epistemic levels.

1. **QFT statements:** Noether currents, trace identities, Callan–Symanzik equations, beta functions, fixed points, stability exponents, matching relations, decoupling, dimensional transmutation, and

vacuum finite-renormalization freedom.

2. **Physical-quotient statements:** which changes preserve the observable or cohomological content, which data transport between scales, and which changes alter the target.
3. **AASC structural-role statements:** anchor, physical action, transport, witness, presentation skin, effective route, obstruction, boundary input, or target update.
4. **Artifact-audit statements:** whether the released source map, tables, encoded examples, theorem ledger, bibliography, and package agree with the manuscript. These are bounded audits, not semantic proofs of QFT or exhaustion.

Exact nonclaim boundary

This manuscript does not derive Standard Model beta functions or anomalous dimensions from AASC; determine numerical couplings, masses, thresholds, or transmutation scales; prove the observed Standard Model scale or conformally invariant; prove unrestricted scale-to-conformal enhancement; solve the hierarchy problem or the cosmological-constant radiative-stability problem; render every cutoff, matching scale, vev, or vacuum quantity unphysical; extend the vacuum-sector correspondence beyond its stated homogeneous scalar target; classify every fixed point or emergent symmetry; treat RG flow as an ordinary internal symmetry on one fixed Hilbert space; or discharge the candidate-completeness and exhaustion theorem reserved for SYM-11.

One-sentence falsifier

The principal theorem fails if, at a fixed target fingerprint, the trace decomposition, quotient descent, scheme transport, universality data, fixed-point classification, EFT matching, transmutation relation, emergence packet, scale-anchor boundary, or vacuum-sector quotient is wrong; if a claimed presentation change alters an admitted physical witness; or if the standing/quotient equivalence is asserted outside the restricted sector in which its bidirectional proof applies. A different field content, operator basis, EFT truncation, background, state, threshold, fixed-point premise bundle, or vacuum sector is a target update unless a transport theorem preserves the complete fingerprint.

Contents

Abstract	i
Reader's guide: theorem spine, claim hierarchy, and exact endpoint	1
I Scope, proof authority, and the scale-language problem	6
1 Arc position and the scale-language problem	6
1.1 Why the word "scale" is structurally ambiguous	6
1.2 Position relative to the preceding SYM modules	6
1.3 The strengthened endpoint	6
2 Fixed targets, controlled fibers, and the scope firewall	7
2.1 The core renormalized-QFT target	7
2.2 Controlled fibers	8
2.3 Three acting loci	8
2.4 Target update versus lawful redescription	8
2.5 Exact nonclaims	9

3	Proof authority and frozen dependency contracts	9
3.1	Authority classes	9
3.2	AASC scale and parameter imports	9
3.3	The zero-point-energy bridge imports	10
3.4	Primary-QFT burdens retained locally	10
4	Parameter roles and mobility modes	10
4.1	Why “constant” and “running” are insufficient types	10
4.2	Worked classification	11
II	Dilatation, trace identities, and quotient RG transport	12
5	Classical dilatation and the improved stress tensor	12
5.1	Active and passive scale transformations	12
5.2	Improvement	12
5.3	Classical Yang–Mills as a diagnostic example	13
6	Renormalized trace identity and anomaly ancestry	13
6.1	Operator rather than slogan	13
6.2	Raw and quotient beta vectors	14
6.3	Gauge-theory specialization	14
6.4	Classical versus quantum scale status	15
6.5	Curved-space boundary	15
7	Callan–Symanzik equations and scale-indexed descriptions	15
7.1	Bare independence and renormalized transport	15
7.2	Description objects	16
7.3	RG time and vector fields	16
8	Physical quotient and RG transport	17
8.1	Descent criterion	17
8.2	Standing-preserving interpretation	17
9	Scheme, regulator, and field-coordinate covariance	18
9.1	Scale-independent local reparameterizations	18
9.2	Explicit one-coupling check	19
9.3	Regulators and subtraction scales	19
III	Universality, fixed points, effective envelopes, and emergence	20
10	Universality quotients	20
10.1	Universality is target-relative	20
10.2	Wilson–Fisher illustration	21
11	Fixed points and conformal status	21
11.1	Coordinate-free fixed loci	21
11.2	Scale invariance and the virial boundary	21
11.3	Irreversibility and endpoint status	22
12	Relevant, irrelevant, marginal, and redundant directions	22

12.1	Redundant operators	23
12.2	Critical exponents and operator dimensions	23
13	Effective envelopes and threshold matching	23
13.1	Different envelopes, one controlled handoff	23
13.2	Tree-level heavy-scalar example	24
13.3	Boundary-trace interpretation	24
14	Dimensional transmutation and RG-invariant scale relations	25
14.1	From a running coordinate to an invariant relation	25
14.2	Role conversion, not creation ex nihilo	25
14.3	Coleman–Weinberg qualification	25
15	IR symmetry and emergence ancestry	26
15.1	IR physicality is not UV provenance	26
15.2	Fixed-point emergence	26
15.3	Heavy-quark symmetry	27
15.4	Anomaly inheritance	27
16	Scale and normalization no-promotion	27
16.1	Scale anchors	27
16.2	What the theorem does not demote	28
16.3	Hierarchy comparisons	28
IV	Realized vacuum-sector correspondence	28
17	A realized-QFT sector: homogeneous vacuum energy and finite renormalization	28
17.1	The restricted sector	28
17.2	Relational and response quantities	29
17.3	Semiclassical gravity and the split degeneracy	30
17.4	What the theorem removes and what it leaves	30
18	Standing and renormalization-quotient descent in the vacuum sector	30
18.1	Binding map	30
18.2	Consequences	32
V	Structural integration, normal form, and handoff	32
19	AASC role map after the QFT classification	32
19.1	Representative, invariant, and target-bearing objects	34
20	The renormalization-covariant scale normal form	34
20.1	Packet definition	34
20.2	Compatibility interfaces	35
21	Rivals, counterexamples, SYM-11 handoff, and conclusion	37
21.1	Rival disposition	37
21.2	Scale-sector handoff to the capstone	37
21.3	Conclusion	38

VI	Reference and reproducibility appendices	38
A	Notation and convention dictionary	38
A.1	Sign conventions	39
A.2	Trace-equality levels	40
B	Selected trace, scheme, and quotient derivations	40
B.1	Local scale Ward identity	40
B.2	Why the beta vector, not its coordinate list, is geometric	40
B.3	Operator coefficients as dual coordinates	40
B.4	Descent proof as a quotient universal property	41
B.5	Vacuum-shift arithmetic	41
B.6	Further explicit checks	41
B.7	Scheme transformation through fourth order	41
B.8	Similarity of the stability matrix	41
B.9	Wilson–Fisher fixed point	42
B.10	Dimensional-transmutation invariant	42
B.11	Tree-level heavy-field elimination	42
B.12	Translation quotient in V_0	42
C	Worked example matrix	43
C.1	Representative versus invariant checklist	43
C.2	Detailed fixed-point, transmutation, and EFT examples	43
C.3	Wilson–Fisher fixed point	44
C.4	Asymptotic freedom and the transmutation relation	44
C.5	Banks–Zaks as a controlled fixed-point example	44
C.6	Heavy-field decoupling and matching	44
C.7	Heavy-quark symmetry as EFT emergence	44
D	Atlas, target, theorem, equation, and source registries	45
D.1	Stable theorem registry	45
D.2	Frozen target family	47
D.3	Source-provenance ledger	47
D.4	Dependency contracts	48
D.5	SYM-10 atlas projection	49
D.6	Parameter role and mobility projection	50
D.7	Equation registry	50
D.8	Bounded rival ledger	50
E	Exact nonclaims and theorem boundary	51
E.1	Boundary interpretation	51
F	Reproducibility and bounded certification	52
F.1	Release contents	52
F.2	Certification doctrine	52
F.3	Build	52
G	Bounded hostile-audit registry	52
G.1	What a hostile scientific objection must do	53
G.2	Verdict rule	53
	References	53

I

Scope, proof authority, and the scale-language problem

1 Arc position and the scale-language problem

1.1 Why the word “scale” is structurally ambiguous

The same word is used for at least five different objects. A physical dilatation transforms spacetime coordinates and fields. A renormalization-group trajectory transports couplings and composite operators among scale-indexed descriptions. A change of subtraction prescription or field normalization is a redescription of the same renormalized theory when observables are transported coherently. An EFT threshold changes the active envelope and operator basis. A fixed point or infrared limit can carry a new physical symmetry that was absent or explicitly broken in the ultraviolet.

Treating these as one operation creates familiar category errors. A running coupling is then mistaken for a freely variable constant; a scheme-dependent beta coefficient is mistaken for an observable; a vanishing beta function is treated as sufficient for conformal invariance; an arbitrary renormalization scale becomes a candidate physical scale; or an effective infrared symmetry is promoted into a primitive ultraviolet principle. The purpose of SYM-10 is to keep the relations among these structures while preventing their collapse.

The distinction is not an anti-RG thesis. It is the condition under which the RG acquires a precise physical role. Callan and Symanzik showed that renormalized correlators obey scale-evolution equations controlled by beta functions and anomalous dimensions rather than by naive engineering dimensions alone [1, 2, 3]. Wilsonian analysis then recast scale evolution as motion in a space of actions and operators, with fixed points and relevant directions organizing universality [4, 5]. Those results are preserved here. The question is what their invariant object is after scheme, field-coordinate, and EFT-envelope changes are admitted.

1.2 Position relative to the preceding SYM modules

SYM-0 supplies the typed symmetry record, quotient-action distinction, exactness lattice, and target-update discipline [6]. SYM-2 supplies the active/passive spacetime distinction and the premise discipline for stress-tensor and Noether statements [7]. SYM-3 separates explicit deformation, global-current anomaly, gauge inconsistency, and matching obstruction [8]. SYM-5 supplies basis, spurion, and invariant-operator discipline [9]. SYM-6 exports target-indexed effective and emergent symmetry examples, including heavy-quark and infrared symmetries [10]. SYM-7 supplies operator-dimension and EFT-order indexing [11]. SYM-9 fixes the rule that RG and emergence claims involving extended operators must preserve the chosen global completion and operator category [12].

None of those imports determines a beta function, proves an interacting fixed point, or computes a threshold coefficient. They constrain the target and the status fields under which such results may be imported or proved.

1.3 The strengthened endpoint

The original arc plan asks SYM-10 to distinguish classical scale symmetry, trace anomaly, RG covariance, fixed points, dimensional transmutation, EFT matching, and emergence. The present manuscript strengthens that target in two ways.

First, it proves a quotient-transport normal form. At each scale μ , one considers a renormalized description containing couplings, masses, sources, a composite-operator basis, matching data, and

a reconstruction map to physical observables. Lawful scheme, field, normalization, and redundant-operator changes define a presentation equivalence relation. When the RG map respects that relation, it descends to a map

$$\overline{\mathcal{R}}_{\mu \rightarrow \nu} : \text{Phys}_{\mu}(D_{10}) \longrightarrow \text{Phys}_{\nu}(D_{10}),$$

with identity and composition on its domain. Physical sameness along the flow is therefore not raw equality of $g^i(\mu)$ and $g^i(\nu)$ but membership in one continuation structure carrying the declared witness ledger.

Second, the manuscript imports a restricted realized-QFT theorem. In the homogeneous metric-proportional vacuum sector, the standard finite-renormalization ambiguity acts transitively on the absolute matter-sector coefficient. Scalar predictive content descends precisely when it is invariant under that quotient. The supplied AASC bridge proves that, under the same restriction, this quotient criterion is equivalent to standing. This is not a general theorem about the full curved-spacetime stress tensor; it is a sharp realized correspondence in one nontrivial sector [13, 14, 15].

Physics translation

The paper's central claim is not that "running is unreal." Running is a lawful physical transport relation when its source and target descriptions, observables, matching data, and quotient are specified. What is rejected is the inference that a running coordinate is an independently reassignable same-domain object, or that a presentation coordinate alone carries the invariant physical identity.

2 Fixed targets, controlled fibers, and the scope firewall

2.1 The core renormalized-QFT target

Definition 2.1 (Scale target). A *scale target* is the tuple

$$D_{10} = (\mathcal{T}_R, \mathcal{O}_R, \mathcal{Q}_{\text{phys}}, I_{\mu}, \mathfrak{S}, \mathfrak{E}, \mathfrak{M}, \mathfrak{B}, \mathfrak{V}, \mathfrak{G}, \mathfrak{A}),$$

with the following fields.

1. $\mathcal{T}_R$ fixes the renormalized QFT or EFT, spacetime dimension, field content, gauge/global structure, interaction basis, quantum regime, and state or phase.
2. $\mathcal{O}_R$ fixes the renormalized composite-operator basis, mixing conventions, equations-of-motion and redundant-operator policy, stress-tensor improvement data, sources, and admitted observables.
3. $\mathcal{Q}_{\text{phys}}$ fixes the gauge, BRST, basis, field-redefinition, and observable quotient appropriate to the target.
4. I_{μ} is the connected scale interval or scale-indexed family over which a continuation claim is made.
5. $\mathfrak{S}$ fixes the class of renormalization schemes, regulators, subtraction prescriptions, field normalizations, and coordinate changes treated as lawful presentations.
6. $\mathfrak{E}$ fixes the EFT envelope, power counting, truncation order, and error domain.
7. $\mathfrak{M}$ fixes thresholds, matching scales, decoupling assumptions, matching maps, and inherited anomaly and symmetry data.
8. $\mathfrak{B}$ fixes background geometry, boundary conditions, external sources, and curvature or improvement data.
9. $\mathfrak{V}$ fixes the vacuum, thermal state, phase, fixed-point target, or IR regime in which symmetry realization is claimed.
10. $\mathfrak{G}$ fixes the symmetry or action data whose exact, approximate, emergent, or absent status is being assessed.
11. $\mathfrak{A}$ fixes anomaly, Ward-identity, matching, and premise data for scale/conformal enhancement, universality, emergence, or the restricted vacuum correspondence.

Target fingerprint

Every theorem in the manuscript is indexed by a target of [theorem 2.1](#). A symbol such as $g(\mu)$, T^μ_μ , Λ_{RG} , “fixed point,” “emergent symmetry,” or “vacuum energy” is not a complete theorem object until the relevant target fields are supplied.

2.2 Controlled fibers

The core comparison family contains several fibers that must not be silently identified.

Fiber	Additional target content
D_{cl}	Classical mass/coupling limit, boundary data, field weights, virial current, and stress-tensor improvement.
D_{RG}	Running renormalized theory over I_μ with operator mixing and observable reconstruction.
D_*	Fixed-point target with declared locality, unitarity, spectrum, stress-tensor, vacuum, and virial-current assumptions.
$D_{\text{EFT}}^{\text{UV/IR}}$	Matched effective envelopes, active field content, operator bases, truncation orders, and overlap observables.
D_{IR}	Standalone IR-symmetry target with a scale window or fixed locus, operator action, state or phase, controlled breaking, and intrinsic anomaly data; no UV completion is required for physical standing.
$D_{\text{UV} \rightarrow \text{IR}}^{\text{em}}$	Emergence-ancestry target adding a specified UV theory, UV-to-IR matching map, deformation ancestry, and anomaly or robust-charge transport.
V_0	Homogeneous metric-proportional vacuum scalar sector with the non-vacuum normalizations and counterterms fixed independently.

A result true in D_* is not automatically true on a running trajectory. A symmetry of an EFT limit is not a symmetry of the complete UV theory. A finite-renormalization statement in V_0 is not a classification of every local curvature tensor in semiclassical gravity.

2.3 Three acting loci

Definition 2.2 (Scale-action locus). The scale-action locus records the object receiving the transformation. The principal loci are:

1. spacetime, fields, states, and local operators under an active dilatation;
2. couplings, masses, sources, and operator coordinates under RG transport;
3. scheme, regulator, subtraction, normalization, or field coordinates under presentation change;
4. field content and operator basis under an EFT-envelope transition;
5. the state, phase, or background under a physical target change.

A single formal rescaling can induce formulas in more than one locus, but the status of the formulas is not inherited automatically. The transformation $x \mapsto e^\sigma x$ can be an active spacetime symmetry at a fixed point; a change $\mu \mapsto e^{-\sigma} \mu$ is a compensating coordinate action in a Callan–Symanzik identity; and a change of units can rescale every dimensionful number while leaving all physical ratios and observables unchanged.

2.4 Target update versus lawful redescription

A proposal changes the target when it changes field content, dimension, background, state, operator category, EFT truncation, threshold content, anomaly structure, fixed-point premise bundle, observable

ledger, or the restricted vacuum sector. A proposal is a lawful redescription only when it carries all dependent data and preserves the admitted physical witnesses.

This distinction is fail-closed for transport claims. When source and target descriptions do not share an overlap witness, when a matching map required for a UV-to-IR emergence claim is absent, when operator mixing is ignored, or when an IR symmetry is asserted outside its scale window, continuation is not inferred from notation alone. The physical standing of a self-contained IR symmetry is assessed within its IR target and does not require an independently specified UV completion.

2.5 Exact nonclaims

The exact nonclaims stated in the reader's guide are part of the target. In particular, numerical boundary conditions, matching inputs, measured scales, and empirical vacuum data may be physical without being internally derived. The manuscript distinguishes *physical* from *internally selected*; it does not equate boundary input with convention or skin.

3 Proof authority and frozen dependency contracts

3.1 Authority classes

The manuscript uses four authority classes.

1. **Local QFT derivation:** the manuscript derives the stated identity or implication directly.
2. **Primary-QFT import:** a standard theorem is imported with its assumptions and scope visible.
3. **AASC structural import:** a fixed role, quotient, continuation, or no-promotion interface is imported only after the corresponding QFT object has been identified.
4. **Bounded artifact audit:** machine-readable records check encoded algebra, source wiring, theorem identity, synchronization, and reproducibility. They do not supply the physical theorem.

The proof order is

QFT statement $\longrightarrow$ physical quotient or observable certificate $\longrightarrow$ AASC role assignment.

No role label is used as the premise proving the QFT statement it later classifies.

3.2 AASC scale and parameter imports

The role-exhaustion theorem for constant-like quantities supplies five role classes: admissibility-fixed, quotient-dependent, effective, transport-defined, and presentation-dependent [16]. The companion mobility analysis separates presentation change, effective drift, transport running, quotient stability, and anchor non-runnability [17]. These are classificatory interfaces. They do not derive the Callan–Symanzik equation or a beta function.

The standing-fixity result supplies the fixed-domain comparison discipline and states explicitly that RG running is compatible with fixity when it is a licensed continuation rather than an arbitrary same-domain reassignment [18]. The RG-standing manuscript supplies the AASC continuation vocabulary for scale-indexed descriptions, overlap witnesses, quotient factorization, universality, fixed points, EFT envelopes, threshold matching, and dimensional transmutation [19]. The continuation-locus and boundary-trace manuscripts supply the structural meaning of persistence through change and of role fixation at a regime boundary [20, 21]. They do not determine the physical matching coefficient or decoupling theorem.

The gauge-coupling, transport-closure, scale-matching, scale-exactness, and hierarchy manuscripts supply concrete no-promotion controls: a running chart coordinate is not a free constant; transport closure is not absolute scale fixation; cross-envelope inputs are not internally derived; and cutoffs, naturalness criteria, fitted vevs, or Planck labels do not acquire scale authority by display alone [22, 23, 24, 25, 26, 27].

3.3 The zero-point-energy bridge imports

The ZPE support stack is used in strict order.

1. The physics-native theorem defines the finite-renormalization quotient in the homogeneous vacuum sector and proves that no nonconstant scalar of the absolute offset descends, while relational differences and the total gravitational coefficient survive in separate roles [13].
2. The physics-bound projection maps that result to the AASC standing, witness, skin, relational, and boundary-input vocabulary [14].
3. The bridge manuscript proves the bidirectional equivalence between standing and quotient descent in the restricted sector [15].

The restriction to V_0 is load-bearing.

3.4 Primary-QFT burdens retained locally

SYM-10 establishes or imports from primary QFT the following science: Noether dilatations and improved stress tensors [28, 29, 30]; Callan–Symanzik equations and renormalized trace identities [1, 2, 31, 32, 33, 34]; Wilsonian universality and fixed-point analysis [4, 5, 35, 36]; asymptotic freedom and dimensional transmutation [37, 38, 39]; decoupling and EFT matching [40, 41, 42]; conditional scale-to-conformal enhancement and the raw-beta/quotient-flow distinction on recurrent conformal trajectories [43, 44, 45, 46]; and the renormalized stress tensor in curved spacetime [47, 48, 49, 50].

Imported interface
An imported primary theorem is not weakened by being imported, and an AASC role is not strengthened into a QFT proof by repetition. The source ledger records which proposition is local, imported, conditional, or structural. A hostile objection must attack the appropriate authority class.

4 Parameter roles and mobility modes

4.1 Why “constant” and “running” are insufficient types

A Lagrangian may contain a gauge coupling, mass, theta angle, Wilson coefficient, subtraction scale, field normalization, matching threshold, or measured boundary value. All can be written as numbers. That syntactic fact does not make them one physical kind. The five-role classification of constant-like quantities and the five mobility modes imported above provide the minimum typed vocabulary required before a scale claim can be assessed [16, 17].

Definition 4.1 (Role profile). For a constant-like quantity C in a fixed target, the role profile $\text{Prof}_{D_{10}}(C)$ is the finite set of loci at which C is:

{admissibility-fixed, quotient-dependent, effective, transport-defined, presentation-dependent}.

A composite symbol can have more than one role only when the roles apply to different components or loci.

Definition 4.2 (Mobility profile). The mobility profile $\text{Mob}_{D_{10}}(C)$ records whether a displayed change is:

{presentation change, effective drift, transport running, quotient stability, anchor non-runability}.

Theorem 4.3 (SYM10-F01: target and locus separation theorem). *For every scale-related claim, the target must specify whether the relevant operation is an active spacetime dilatation, passive coordinate/unit/normalization change, RG transport in coupling and operator space, scheme or field-coordinate redescription, EFT-envelope transition, or change of physical state, phase, background, or target. No conclusion passes from one locus to another without an explicit theorem or matching map preserving the target data required by the conclusion.*

Proof. The loci act on different objects and possess different tests. A physical dilatation is diagnosed by a current, Ward identity, and action on states or operators. RG transport is diagnosed by scale-indexed correlation functions, beta functions, anomalous dimensions, and reconstruction maps. A scheme change is an invertible coordinate change preserving the physical quotient only when dependent data are transported. An EFT transition changes the active field and operator envelope and requires matching. A state or phase change alters physical expectation values. Identifying any two without a theorem makes their source, target, and witness maps ill typed. $\square$

Theorem 4.4 (Parameter-role and mobility normal form (SYM10-F02)). *At fixed target, every admissibility-bearing use of a constant-like quantity decomposes into the role and mobility profiles above. Running is not arbitrary numerical reassignment: it is transport-defined mobility on a scale-indexed continuation locus. A change of an anchor changes the target unless a separate theorem identifies the change as lawful transport of the complete anchor-bearing structure.*

Proof. The role theorem follows from the five loci at which a constant-like quantity can contribute: target eligibility, quotient-level invariant content, envelope-bounded validity, lawful transport, and representation [16]. The mobility theorem separates changes that preserve the physical quotient, changes induced by an effective envelope, changes generated by a transport law, invariant content preserved through the change, and target-bearing data that cannot move without changing the comparison class [17, 18]. The classes are exhaustive relative to the frozen target because any displayed change either preserves the quotient, is part of the scale or envelope evolution, or changes a target field. Cross-role substitution fails: a scheme coordinate cannot select an anchor, and a boundary input is not converted into an internal derivation by stable transport. $\square$

4.2 Worked classification

Quantity	Primary role/mobility	Blocked inference
$g(\mu)$	Transport-defined; RG running	The value at one μ is not a freely chosen same-domain coupling once the trajectory and boundary data are fixed.
$\beta(g)$	Transport vector field; scheme-covariant	A raw higher-loop coefficient is not automatically an observable.
μ	Presentation/transport coordinate	Choosing μ does not select a physical mass scale.
$C_A(\mu)$	Effective and transport-defined	A Wilson coefficient outside its EFT and truncation is not a universal constant.
Λ_{RG}	Quotient-stable transmutation invariant	RG invariance does not numerically predict the scale without boundary or matching data.
μ_m	Matching coordinate; sometimes empirical boundary input	The matching scale is not a new particle threshold unless tied to a physical mass or witness.
Critical exponent	Quotient-dependent universal datum	A coordinate eigenvalue before redundant directions are removed need not be physical.
ρ_R in V_0	Presentation-dependent homogeneous offset	The regulator sum is not a standalone cosmological prediction.
Λ_{eff}	Matched total gravitational coefficient	Its physicality does not identify the split into matter and gravitational counterterms.

Adversarial check

Dimensionless quantities are not automatically more primitive than dimensional ones. A dimensionless coupling can be scheme-dependent and transport-defined; a dimensional pole mass can be an observable; a dimensionless ratio can remain unanchored if no absolute scale certificate is supplied. Role precedes explanatory use.

II**Dilatation, trace identities, and quotient RG transport****5 Classical dilatation and the improved stress tensor****5.1 Active and passive scale transformations**

Consider a classical field theory on d -dimensional Minkowski space with fields ϕ_A of engineering dimensions Δ_A . An infinitesimal active dilatation acts as

$$\delta_\sigma x^\mu = 0, \quad \delta_\sigma \phi_A(x) = -\sigma(x \cdot \partial + \Delta_A)\phi_A(x),$$

when written at fixed coordinate argument. Equivalently, one can transform coordinates and compare the pulled-back field. The passive coordinate formula is not itself a new physical action; the active transformation is the object relevant to a Noether charge and to its action on states or operators [28, 29].

If the action changes by a boundary term,

$$\delta_\sigma S = \sigma \int d^d x \partial_\mu K^\mu,$$

Noether's theorem yields a dilatation current

$$D^\mu = x_\nu T^{\mu\nu} - V^\mu,$$

where V^μ is the virial current. On the equations of motion,

$$\partial_\mu D^\mu = T^\mu{}_\mu - \partial_\mu V^\mu.$$

Thus engineering covariance of the Lagrangian is not sufficient by itself. The current, its boundary flux, the stress tensor, and the virial term must all be well-defined at the claimed target.

5.2 Improvement

The stress tensor is not unique. If a local tensor $L^{\rho\mu\sigma\nu}$ has the algebraic symmetries required for an improvement, then

$$T^{\mu\nu} \mapsto T^{\mu\nu} + \partial_\rho \partial_\sigma L^{\rho\mu\sigma\nu}$$

preserves conservation and the Poincaré charges under suitable boundary conditions. For a free massless scalar,

$$\mathcal{L} = \frac{1}{2} \partial_\mu \phi \partial^\mu \phi,$$

the conformal improvement is

$$\Theta_{\mu\nu} = T_{\mu\nu} + \xi_c (\eta_{\mu\nu} \square - \partial_\mu \partial_\nu) \phi^2, \quad \xi_c = \frac{d-2}{4(d-1)}.$$

Using $\square\phi = 0$, one obtains $\Theta^\mu{}_\mu = 0$ in flat space. The improvement is not a numerical selection of a new physical scale. It is a change within the admissible stress-tensor representative class, provided the boundary and curvature couplings are transported coherently [29, 30].

Theorem 5.1 (Classical scale-action theorem (SYM10-Q01)). *A classical target carries a physical dilatation structure only when the following data are fixed and satisfied:*

1. *all dimensionful explicit deformations relevant to the claim are absent or transformed as declared spurions;*
2. *the couplings occupy a classically scale-covariant interaction basis;*
3. *the background and boundary conditions admit the corresponding current and charge;*
4. *the virial current and improvement class are specified;*
5. *the transformation acts nontrivially on the admitted state or operator quotient when a physical symmetry, rather than mere covariance, is claimed.*

Engineering-dimension covariance alone does not establish a conserved physical dilatation charge.

Proof. The Noether identity gives $\partial_\mu D^\mu = T^\mu{}_\mu - \partial_\mu V^\mu$. A conserved current therefore requires vanishing of this combination at the stated level: pointwise on shell, as an integrated charge under boundary conditions, or modulo improvements and equations of motion. An explicit mass term contributes a nonzero trace and breaks the symmetry unless treated as a transforming source. A passive coordinate or unit change can preserve all physical data without producing a charge. Conversely, when the current exists and its charge acts on states or operators, the action descends nontrivially to the physical quotient. The listed data are exactly the distinctions required to tell these cases apart. $\square$

5.3 Classical Yang–Mills as a diagnostic example

In four dimensions, the classical Yang–Mills action

$$S = -\frac{1}{4g^2} \int d^4x F_{\mu\nu}^a F^{a\mu\nu}$$

contains a dimensionless coupling and is classically scale invariant on flat space. The improved stress tensor has vanishing trace modulo equations of motion. This statement does not survive quantization unchanged: the running of g yields a beta-function term in the renormalized trace. The example will be used in [section 6](#) to separate classical exactness from quantum anomaly.

Adversarial check

A classical trace that can be improved to zero does not prove quantum conformal invariance. A vanishing integrated trace does not prove a local operator identity. A passive rescaling of coordinates does not prove a physical symmetry. These are distinct targets and certificate levels.

6 Renormalized trace identity and anomaly ancestry

6.1 Operator rather than slogan

The trace of the renormalized energy-momentum tensor is a composite-operator statement. Its exact form depends on the interaction basis, operator mixing, improvement convention, background sources, and whether the equality is asserted as a local operator relation, inside separated-point correlators, or modulo equations of motion and contact terms. The word “anomaly” does not remove those dependencies.

Let g^i denote dimensionless couplings multiplying renormalized operators $[O_i]$, and let m^a denote relevant mass coordinates. In a flat-space basis adapted to the target, the trace has the schematic form

$$[T^\mu{}_\mu] = \sum_i \beta^i(g) [O_i] + \sum_a (1 + \gamma_{m_a}) m^a [O_a] + \partial_\mu V^\mu + \mathcal{I}_{\text{impr}} + \mathcal{E}_{\text{EOM/contact}}. \quad (1)$$

Curved backgrounds add local curvature terms and Weyl anomalies. Source-dependent local RG also adds derivatives of couplings and background-field terms [33, 34].

6.2 Raw and quotient beta vectors

The coordinate beta vector need not itself be the physical fixed-point criterion when the coupling description contains flavor or redundant directions. Let

$$\pi : \mathcal{M}_{\text{coupl}} \longrightarrow \mathcal{M}_{\text{phys}}$$

denote the quotient by the admitted redundant and flavor-rotation directions. The induced physical flow is

$$\bar{\beta} := \pi_* \beta. \quad (2)$$

In local-RG conventions where a generator S represents the vertical flavor rotation, a convenient representative of the same quotient flow is

$$\mathcal{B}^I = \beta^I - (Sg)^I, \quad (3)$$

with the sign tied to the convention for S . The invariant statement is (2): a physical fixed locus has $\bar{\beta} = 0$. Raw coordinate beta functions can remain nonzero along a recurrent flavor or redundant orbit even when the quotient theory is stationary. This distinction is required by the local-RG analysis of recurrent conformal trajectories [51, 33, 45].

Accordingly, the beta contribution to a physical trace statement is the nonredundant quotient contribution after flavor-current or virial terms have been combined consistently. Equation (1) remains a valid schematic coordinate representation, but a raw component β^I must not be interpreted as an invariant measure of physical scale breaking in isolation.

6.3 Gauge-theory specialization

For a renormalized gauge theory with canonical kinetic normalization, a common convention is

$$[T^\mu_\mu] = \frac{\beta(g)}{2g} [F^a_{\rho\sigma} F^{a\rho\sigma}] + (1 + \gamma_m) m [\bar{\psi}\psi] + \cdots,$$

where the omitted terms include gauge-fixing/BRST-exact, EOM, mixing, and background contributions. The numerical coefficient in front of $[F^2]$ changes if the coupling is placed outside the kinetic term in a different way. The invariant content is the complete operator identity and its physical matrix elements, not a convention-free isolated coefficient [31, 32, 52].

Theorem 6.1 (Renormalized trace-identity theorem (SYM10-Q02)). *Fix a renormalized target D_{10} with operator basis, mixing convention, stress-tensor improvement, background, and source ledger. Then the renormalized trace decomposes into separately typed contributions:*

1. *explicit relevant deformations;*
2. *nonredundant beta-function terms associated with quantum scale dependence, with raw flavor or redundant motion kept separately typed;*
3. *anomalous-dimension and operator-mixing terms;*
4. *virial-current and improvement terms;*
5. *equations-of-motion and contact terms;*
6. *curvature and background-source contributions where admitted.*

A lawful scheme or operator-basis change may alter the coordinate representation of the decomposition, but it must preserve the quotient beta flow, the complete correlator or observable content, and the anomaly/matching class.

Proof. The local Callan–Symanzik equation treats couplings and masses as sources. When flavor or redundant directions are present, its physically stationary content is expressed by the induced quotient flow $\bar{\beta}$ (or an equivalent local representative such as $\mathcal{B}$), not by componentwise vanishing of every raw beta coordinate. An infinitesimal Weyl rescaling can be compensated by beta-function and anomalous-dimension variations of those sources, leaving a local anomaly functional. Differentiation with respect to the metric yields the trace identity; differentiation with respect to operator sources gives the corresponding correlator identities [33, 34]. Explicit masses appear because their classical dimensions and anomalous dimensions obstruct scale invariance. Operator mixing and EOM/contact terms arise because the renormalized local basis is not diagonal or unique. Improvement shifts the representative stress tensor without changing the conserved Poincaré generators under the declared boundary conditions. Scheme changes push forward the beta functions and mix the operators, preserving the full functional identity. The listed terms therefore form distinct causal and epistemic classes even when several appear in one displayed trace. $\square$

6.4 Classical versus quantum scale status

For massless Yang–Mills, $T^\mu{}_\mu = 0$ classically in four dimensions, while quantization yields

$$[T^\mu{}_\mu] = \frac{\beta(g)}{2g}[F^2] + \mathcal{E}_{\text{EOM/BRST/contact}}.$$

The beta term is not an ordinary explicit mass deformation. It records quantum scale dependence of the nonredundant transport. At a physical fixed locus the induced quotient beta $\bar{\beta}$ vanishes, so the nonredundant beta contribution disappears; raw beta components may remain along flavor or redundant directions and are then paired with the corresponding current or presentation terms. Conformal enhancement still depends on the virial current, stress tensor, state, spectrum, and unitarity/locality assumptions discussed later.

6.5 Curved-space boundary

On a curved background, the trace includes local curvature invariants and scheme-dependent total derivatives. Wald’s axiomatic analysis and the later local-covariant construction of Wick polynomials show that finite renormalization of the stress tensor includes conserved local curvature tensors [47, 48, 49]. The vacuum-sector theorem in Part IV deliberately freezes those terms and isolates the homogeneous metric-proportional scalar ambiguity. Nothing in that restricted theorem removes the curvature-dependent trace anomaly.

Exact nonclaim boundary

The trace identity does not say that every nonzero trace is an anomaly, that every beta function is physical coordinate-by-coordinate, that every improvement is skin on arbitrary boundaries, or that a vanishing beta function alone establishes conformal symmetry.

7 Callan–Symanzik equations and scale-indexed descriptions

7.1 Bare independence and renormalized transport

A renormalized correlation function depends on the subtraction scale because the renormalized coordinates used to represent one bare theory depend on that scale. Independence of the bare description gives the Callan–Symanzik equation. For a renormalized n -point function, a convenient convention is

$$\left(\mu \frac{\partial}{\partial \mu} + \beta^i(g) \frac{\partial}{\partial g^i} + \gamma_{m_a} m^a \frac{\partial}{\partial m^a} + n \gamma_\phi \right) G_R^{(n)} = 0, \quad (4)$$

with sign and multiplicity conventions fixed once for the target. Composite-operator insertions require a matrix anomalous dimension,

$$\mu \frac{d}{d\mu} [O_A] = -\gamma_A^B [O_B] + \mathcal{E}_{\text{EOM/contact}}.$$

The equation is a transport equation on a bundle of descriptions. It does not identify an RG step with a physical spacetime symmetry. Away from a fixed point, the scale derivative is compensated by motion in coupling and operator space. At a physical fixed locus, the induced nonredundant beta vector $\bar{\beta}$ vanishes, although raw beta components may persist along redundant or flavor rotations. The remaining scaling law can become the Ward identity of a physical scale symmetry when the target premises are satisfied [1, 2, 3, 45].

7.2 Description objects

Definition 7.1 (Scale-indexed description). A scale-indexed description $d_\mu \in \text{Desc}_\mu(D_{10})$ is a tuple

$$d_\mu = (g^i(\mu), m^a(\mu), C_A(\mu), [O_A]_\mu, Z_\mu, \text{extsources}, \mathcal{W}_\mu),$$

where $\mathcal{W}_\mu$ is the reconstruction map to the admitted observables, amplitudes, correlators, or state data.

A description is more than a list of couplings. Two coordinate lists related by a field redefinition are not known to describe the same physics until the operator basis, sources, Jacobians or anomaly data, matching maps, and reconstruction functions are transported.

Definition 7.2 (Presentation equivalence). Two descriptions d_μ, d'_μ are presentation-equivalent, written $d_\mu \sim_{\text{pres}} d'_\mu$, when an admitted invertible scheme, regulator, field-coordinate, normalization, basis, or redundant-operator transformation carries the complete target data of d_μ to d'_μ and preserves the witness ledger through the reconstructed observables.

The physical quotient at scale μ is

$$\text{Phys}_\mu(D_{10}) = \text{Desc}_\mu(D_{10}) / \sim_{\text{pres}}.$$

This quotient is target-relative. A field redefinition that is lawful in a local operator sector can fail when boundary conditions, extended operators, anomalies, or a truncated EFT basis are changed.

7.3 RG time and vector fields

Write $t = \log(\mu/\mu_0)$. The RG vector field on coupling space is

$$\beta = \beta^i(g) \frac{\partial}{\partial g^i}, \quad \frac{dg^i}{dt} = \beta^i(g).$$

The flow may be partial: Landau poles, thresholds, phase transitions, singular coordinate charts, or the loss of an overlap witness can end the continuation domain. A formal solution of the differential equation beyond such a boundary is not automatically a physical continuation.

Physics translation

The Callan–Symanzik equation expresses scale independence of bare physics as covariance of renormalized descriptions. The running coordinates are real and computationally indispensable. Their role is transport, not arbitrary same-domain multiplicity.

8 Physical quotient and RG transport

8.1 Descent criterion

Let

$$\mathcal{R}_{\mu \rightarrow \nu} : \text{Desc}_{\mu}(D) \longrightarrow \text{Desc}_{\nu}(D)$$

be the RG evolution defined by the beta functions, anomalous dimensions, and operator transport on the interval $[\mu, \nu] \subset I_{\mu}$. To descend to the physical quotient, it must respect presentation equivalence:

$$d \sim_{\text{pres}} d' \implies \mathcal{R}_{\mu \rightarrow \nu}(d) \sim_{\text{pres}} \mathcal{R}_{\mu \rightarrow \nu}(d'). \quad (5)$$

This is the precise form of the claim that scheme or field coordinates do not obstruct physical RG identity.

Theorem 8.1 (SYM10-Q03: RG quotient-transport theorem). *Assume:*

1. $\text{Desc}_{\mu}(D)$ and $\text{Desc}_{\nu}(D)$ reconstruct a common admitted observable class on an overlap domain;
2. $\sim_{\text{pres}}$ is an equivalence relation preserving that observable and state data;
3. $\mathcal{R}_{\mu \rightarrow \nu}$ is well defined on the scale interval and satisfies (5);
4. threshold or envelope boundaries inside the interval are supplied with matching maps satisfying the corresponding overlap conditions.

Then $\mathcal{R}_{\mu \rightarrow \nu}$ descends uniquely to

$$\bar{\mathcal{R}}_{\mu \rightarrow \nu} : \text{Phys}_{\mu}(D) \longrightarrow \text{Phys}_{\nu}(D), \quad \bar{\mathcal{R}}_{\mu \rightarrow \nu}([d]) = [\mathcal{R}_{\mu \rightarrow \nu}(d)].$$

Where the underlying RG maps compose, the quotient maps satisfy

$$\bar{\mathcal{R}}_{\nu \rightarrow \lambda} \circ \bar{\mathcal{R}}_{\mu \rightarrow \nu} = \bar{\mathcal{R}}_{\mu \rightarrow \lambda}.$$

Proof. If $[d] = [d']$, then $d \sim_{\text{pres}} d'$. Equivariance implies $\mathcal{R}(d) \sim_{\text{pres}} \mathcal{R}(d')$, so the image class is independent of the representative. This proves existence and well-definedness. Uniqueness follows because the quotient projection is surjective. Composition follows by applying the quotient projection to the composition of the underlying maps. At a threshold, the same argument applies to the matching map provided it preserves the overlap observables and quotient relation. $\square$

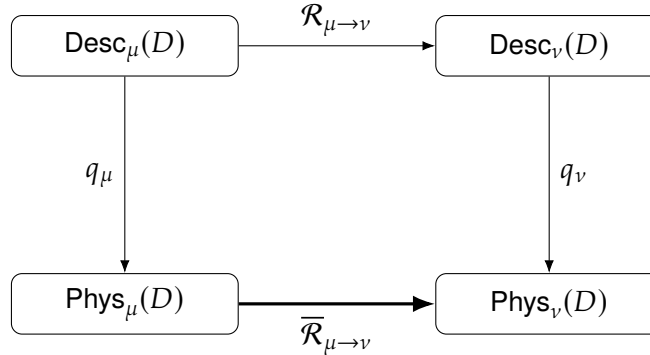


Figure 1: RG transport descends only when the upper map is equivariant with the presentation quotient.

8.2 Standing-preserving interpretation

The imported AASC transport theorem identifies lawful continuation with preservation of a standing-bearing witness through a quotient map [19]. In the present paper, that structural statement is applied only after the QFT quotient has been constructed. A raw family $g(\mu)$ without overlap observables, operator transport, or an admissible interval is not promoted into a physical trajectory by notation alone.

Adversarial check

A discontinuity, Landau pole, threshold without matching, phase transition, or loss of the admitted operator basis blocks the unqualified composition law. The correct response is a target update or matching theorem, not a silent extension of $\overline{\mathcal{R}}$.

9 Scheme, regulator, and field-coordinate covariance

9.1 Scale-independent local reparameterizations

Let $g'^i = f^i(g)$ be an invertible local reparameterization on the admitted coupling patch with no explicit μ dependence. The raw coordinate beta functions transform as a vector field:

$$\beta'^i(g') = \frac{\partial f^i}{\partial g^j} \beta^j(g). \quad (6)$$

This formula is necessary but not sufficient for physical equivalence. Composite operators, Wilson coefficients, anomalous dimensions, sources, matching maps, and observable reconstruction are coupled to the choice of coordinates.

For a transformation with explicit scale or mass-ratio dependence, write the complete dimensionless coordinate list as x^A and

$$x'^A = F^A(x, \mu).$$

Then the chain rule gives

$$\beta'^A = \frac{\partial F^A}{\partial x^B} \beta^B + \mu \left. \frac{\partial F^A}{\partial \mu} \right|_x. \quad (7)$$

A mass-dependent scheme is therefore not governed by (6) alone. The same rule applies on the physical quotient after projecting away admitted redundant or flavor directions; the induced vector field $\bar{\beta}$ is the invariant transport object.

If the renormalized operator basis changes according to

$$[O'_A]_R = M_A^B(g)[O_B]_R + \mathcal{E}_A, \quad (8)$$

where $\mathcal{E}_A$ lies in the declared equation-of-motion or redundant subspace, then an interaction pairing $C^A[O_A]_R$ is preserved only if

$$C'^A = C^B(M^{-1})_B^A \quad (9)$$

modulo those redundant directions. With the convention

$$\mu \frac{d}{d\mu} [O]_R = -\gamma [O]_R,$$

the anomalous-dimension matrix transforms, for a basis $M(g)$ with no additional explicit scale dependence, as

$$\gamma' = M\gamma M^{-1} - (\beta \cdot \partial M)M^{-1}. \quad (10)$$

Explicit scale dependence of M supplies the corresponding additional $-\mu(\partial_\mu M)M^{-1}$ term.

Lemma 9.1 (Normalization and scheme covariance lock). *A scheme, normalization, field, coupling, subtraction, or operator-basis transformation is admitted as a presentation map only if it is invertible on the declared quotient coordinates and transports the complete dependent datum: raw and quotient beta vectors, masses, fields, composite operators, Wilson coefficients, anomalous dimensions, sources, matching maps, boundary conditions, and observable reconstruction. Explicitly scale- or mass-dependent transformations must use the extended chain rule (7). A partial transformation that leaves a conjugate datum fixed is not lawful.*

Proof. Equation (6) is the chain rule for a scale-independent coupling-coordinate map, while (7) is the corresponding rule on the extended scale-dependent coordinate space. Equations (8)–(9) preserve the operator/coefficient pairing. Differentiating a scale-dependent basis change along the RG flow gives (10) together with the explicit-scale term when present. Sources and matching maps are transported on the corresponding dual spaces. Therefore a lawful map preserves the complete generating-functional or observable reconstruction record. Holding any conjugate datum fixed generally changes the physical prediction and fails the presentation criterion. $\square$

Theorem 9.2 (SYM10-Q04: scheme and presentation covariance). *Under a lawful transformation satisfying theorem 9.1, physical correlator content, amplitudes, pole data, anomaly classes, matched predictions, and the quotient RG trajectory are preserved at the declared accuracy. Correlator components transform covariantly under operator or source basis changes. Raw coupling coordinates, raw beta components, fixed-point representatives, higher beta-function coefficients, operator representatives, and subtraction constants need not be invariant.*

Proof. The transformed data satisfy the primed Callan–Symanzik equation because the extended RG vector, operator basis, and anomalous dimensions transform coherently. Projection to the physical quotient commutes with lawful presentation transport. The action and source pairings are unchanged by contragredient transport, and matching maps commute with the source and target basis changes. Therefore the reconstructed physical quantities agree, while coordinate-dependent components can differ. $\square$

9.2 Explicit one-coupling check

Let

$$\beta(g) = b_1 g^2 + b_2 g^3 + b_3 g^4 + \mathcal{O}(g^5), \quad g' = g + a g^2.$$

Applying (6) and re-expanding in g' gives

$$\beta'(g') = b_1 g'^2 + b_2 g'^3 + (b_3 - a b_2 - a^2 b_1) g'^4 + \mathcal{O}(g'^5).$$

The displayed coefficient pattern belongs to this normalization and power convention. In gauge-coupling expansions beginning at g^3 , the familiar universal perturbative coefficients occupy the corresponding first two loop orders.

At a quotient fixed locus choose local transverse coordinates $\bar{g}^\alpha$ on $\mathcal{M}_{\text{phys}}$, so that the induced beta vector $\bar{\beta}^\alpha$ vanishes there. For a scale-independent smooth reparameterization with Jacobian J^α_β , the nonredundant stability matrix obeys

$$B' = J B J^{-1}. \quad (11)$$

The result does not require componentwise vanishing of a raw beta vector along vertical flavor or redundant directions. Thus nonredundant critical exponents are invariant even though the coordinate representative of the fixed class can change.

9.3 Regulators and subtraction scales

A regulator is not automatically skin in every context. Inside a completed renormalized target it can be removed or changed without altering the physical quotient and is then presentational. In an EFT, a cutoff can also mark the boundary of the envelope or the onset of omitted degrees of freedom. A physical threshold is not equivalent to an arbitrary regulator coordinate. Its role follows from the target and matching certificate, not from the symbol Λ .

Certificate

The release artifacts verify selected finite instances of (6), (7), (9), (10), and (11). They do not prove the existence of every scheme transformation, the full Callan–Symanzik theorem, or semantic

completeness of the scheme class.

III

Universality, fixed points, effective envelopes, and emergence

10 Universality quotients

10.1 Universality is target-relative

Universality is often summarized by the statement that many microscopic systems share the same critical exponents. That statement is useful but incomplete. A universality claim can also include the fixed point, relevant operator content, global and generalized symmetries, anomalies, boundary conditions, scaling functions, and state or phase. Two models can share one exponent while differing in operator algebra or symmetry realization.

Definition 10.1 (Universality quotient). Fix an IR target D_* and a declared set of universal data $\mathcal{U}_*$. Two microscopic descriptions X and Y are universality-equivalent when their RG transports enter the basin of the same quotient-stable fixed point and agree on $\mathcal{U}_*$ after microscopic surplus and redundant coordinates are removed.

The quotient can be coarse or fine, but it must be declared. A critical-exponent quotient is weaker than a complete CFT quotient. A bulk universality class can split when boundary conditions or defects are admitted. A local-operator universality class can split when the global form or extended-operator sector fixed in SYM-9 is included. An irrelevant coupling is also not automatically disposable for every observable: a dangerously irrelevant variable can vanish at the fixed point while still controlling scaling amplitudes, order-parameter relations, or neighboring-phase data. Whenever the asserted universal content depends on such a variable, that dependence belongs in $\mathcal{U}_*$ rather than being quotiented away by name alone [53].

Theorem 10.2 (Universality quotient theorem (SYM10-Q05)). *Two microscopic descriptions belong to one universality class at the asserted scope only if their RG transports agree on the fixed target and on every datum claimed universal, including the nonredundant relevant directions, critical exponents, operator and symmetry content, anomaly or matching data, and any stated universal scaling functions. Microscopic differences not retained by this quotient are surplus relative to the target; differences in the declared universal data change the universality target.*

Proof. The Wilsonian RG integrates out short-distance degrees of freedom and rescales, producing trajectories in a space of actions. Irrelevant coordinates decay toward the fixed point, while relevant coordinates must be tuned to reach the critical surface. The resulting long-distance observables depend on the fixed point and the finite set of retained perturbations rather than on every microscopic coupling [4, 5]. This establishes a quotient by microscopic surplus. The exact equivalence relation is determined by the data the theorem claims to preserve. If anomaly, symmetry, boundary, or operator information is included in the conclusion, agreement of exponents alone cannot imply equivalence. Conversely, differences only in irrelevant or redundant coordinates do not define a new universality class at that coarser target. $\square$

10.2 Wilson–Fisher illustration

For the $O(N)$ scalar model in $d = 4 - \epsilon$, choose a dimensionless coupling u with one-loop beta function

$$\beta(u) = -\epsilon u + \frac{N+8}{6}u^2 + \mathcal{O}(u^3).$$

Besides the Gaussian fixed point, the Wilson–Fisher fixed point is

$$u_* = \frac{6\epsilon}{N+8} + \mathcal{O}(\epsilon^2).$$

The thermal scaling exponent is

$$\nu^{-1} = 2 - \frac{N+2}{N+8}\epsilon + \mathcal{O}(\epsilon^2),$$

while the field anomalous dimension begins at order ϵ^2 . Microscopically different lattice or continuum realizations can share these long-distance data, but only after the symmetry, dimensionality, boundary, and operator target is fixed [35, 4, 53].

Physics translation

In the AASC role map, a universality class is not a new microscopic substance. It is a quotient-persistent physical structure: a fixed target plus the data that survive elimination of microscopic surplus.

11 Fixed points and conformal status

11.1 Coordinate-free fixed loci

A fixed point is often written as $\beta^i(g_*) = 0$. That notation is sufficient in a regular chart with no redundant or flavor-rotation direction, but it is not the invariant definition. Let

$$\pi : \mathcal{M}_{\text{coupl}} \rightarrow \mathcal{M}_{\text{phys}}$$

be the quotient fixed in [section 6](#), and let $\bar{\beta} = \pi_*\beta$ be the induced physical RG vector field. A physical fixed locus is a zero of $\bar{\beta}$. Raw coordinate beta functions can remain nonzero when they generate only a vertical flavor or redundant motion; recurrent conformal trajectories provide explicit perturbative examples [51, 33, 45].

Definition 11.1 (Transport-stable fixed point). A transport-stable fixed point is a quotient class $[d_*] \in \text{Phys}(D_{10})$ such that

$$\bar{\beta}([d_*]) = 0,$$

the descended RG transport leaves the class fixed, and the class carries non-null operator, state, and observable data. A raw recurrent trajectory is allowed only when it projects to the same fixed physical class.

11.2 Scale invariance and the virial boundary

At a quotient fixed locus, the nonredundant beta contribution to the flat-space trace vanishes. Raw beta components can remain along flavor or redundant directions and combine with current-divergence terms without representing physical scale evolution. After that quotient/improvement step, the remaining trace can have the form

$$T^\mu{}_\mu = \partial_\mu V^\mu + \mathcal{I}_{\text{impr}} + \mathcal{E}_{\text{EOM/contact}}.$$

A conserved dilatation current exists when the virial term is compatible with the target. Conformal invariance requires the stronger condition that the stress tensor can be improved so that its trace vanishes, or an equivalent condition appropriate to the dimension and assumptions. The scale-to-conformal problem is theorem-dependent: it is established in important classes, including unitary two-dimensional QFTs and broad perturbative four-dimensional settings, but it is not an assumption-free identity in every dimension and every theory [43, 44, 45, 46].

Theorem 11.2 (Fixed-point and conformal-status theorem (SYM10-Q06)). *A physical RG fixed point is a quotient-stable zero of the induced RG vector field $\bar{\beta}$ carrying the declared operator and witness data. Componentwise vanishing of every raw beta coordinate is not required when the residual raw motion is purely redundant or a flavor rotation. At such a locus, scale invariance follows only when the state, boundary, stress-tensor, and virial-current conditions needed for a conserved dilatation current hold. Conformal enhancement is concluded only under an explicit premise bundle sufficient to improve or otherwise control the trace in the relevant dimension and quantum regime.*

Proof. Lawful reparameterizations induce the same quotient RG vector field, so the condition $\bar{\beta} = 0$ is presentation-covariant. The trace identity of theorem 6.1 shows that the nonredundant beta contribution vanishes at the quotient fixed locus while vertical flavor or redundant motion is absorbed into the appropriate current or presentation sector. A fixed class with an ill-defined state, null operator algebra, nontrivial unremovable virial current, or incompatible boundary conditions does not automatically carry the desired physical scale or conformal action. The cited scale-to-conformal theorems establish enhancement only after their assumptions are supplied. Therefore the correct status is conditional and target-indexed. $\square$

Adversarial check

A nonzero raw beta vector is not by itself evidence against conformality when it is tangent to an admitted flavor or redundant orbit. Conversely, a zero of a truncated raw beta function is not by itself a physical fixed point. The invariant question is whether the induced flow on the nonredundant physical quotient vanishes and whether the stress-tensor premise ledger is satisfied.

11.3 Irreversibility and endpoint status

Zamolodchikov's two-dimensional c -theorem and the four-dimensional a -theorem provide monotone quantities under their respective assumptions, constraining RG flows between conformal fixed points [54, 55]. They do not turn every trajectory into a symmetry action or prove that every endpoint is conformal. They are witnesses of irreversibility and endpoint structure at declared targets.

Exact nonclaim boundary

The paper does not claim that the observed Standard Model lies at an interacting fixed point, that a perturbative quotient zero is nonperturbatively exact, or that scale invariance universally implies conformal invariance.

12 Relevant, irrelevant, marginal, and redundant directions

Let $\bar{g}^\alpha$ be local coordinates on the nonredundant physical quotient near a fixed class $[g_*]$, and let $\bar{\beta}^\alpha$ denote the induced quotient RG vector field. Linearization gives

$$\mu \frac{d}{d\mu} \delta \bar{g}^\alpha = B^\alpha{}_\beta \delta \bar{g}^\beta + O(\delta \bar{g}^2), \quad B^\alpha{}_\beta = \left. \frac{\partial \bar{\beta}^\alpha}{\partial \bar{g}^\beta} \right|_{[g_*]}. \quad (12)$$

The signs used to label relevant and irrelevant directions depend on whether the flow parameter is taken toward the infrared or ultraviolet. An eigenperturbation that grows toward the infrared is relevant to the infrared fixed point, one that decays is irrelevant, and one with zero linear eigenvalue is marginal pending higher-order analysis.

12.1 Redundant operators

A perturbation generated by a local field redefinition, an equation-of-motion operator, a pure flavor rotation, or another admitted reparameterization is redundant. Its raw coordinate motion need not correspond to a physically distinct deformation. The stability analysis is therefore performed on

$$T_{[g_*]} \mathcal{M}_{\text{phys}} = T_{g_*} \mathcal{M}_{\text{coupl}} / T_{g_*} \mathcal{M}_{\text{red}}.$$

A recurrent raw trajectory can consequently have a stationary quotient class and a well-defined physical response matrix even though the raw coupling coordinates continue to rotate.

Theorem 12.1 (SYM10-Q07: nonredundant response theorem). *After quotienting redundant and flavor-rotation directions, the spectrum and Jordan structure of the linearized induced RG map on $T_{[g_*]} \mathcal{M}_{\text{phys}}$ are invariant under smooth invertible reparameterization of the quotient coordinates. They classify relevant, irrelevant, and linearly marginal physical responses at the fixed target. Exactly marginal, marginally relevant, and marginally irrelevant directions require the nonlinear quotient beta functions and operator-product data.*

Proof. At a quotient fixed class, $\bar{\beta}([g_*]) = 0$. An invertible change of quotient coordinates has Jacobian J , and differentiating the pushforward law for $\bar{\beta}$ gives

$$B' = JB^{-1}.$$

Thus the nonredundant eigenvalues and Jordan structure are invariant. Redundant and flavor directions must first be removed because field redefinitions or symmetry rotations can otherwise contribute spurious raw eigenvectors. A zero eigenvalue supplies no nonlinear conclusion; higher-order terms decide whether the direction is exactly marginal or slowly runs. $\square$

12.2 Critical exponents and operator dimensions

For a conformal fixed point, the RG eigenvalues are related to scaling dimensions of the corresponding operators. This relation is physical only after operator mixing and redundancy are controlled. A critical exponent is therefore a quotient-level witness, while the matrix entries of a raw stability matrix are coordinate-dependent.

Adversarial check

A family of zeros of a truncated raw beta function is not automatically a conformal manifold, and a recurrent raw trajectory is not automatically a physical cycle. One must identify the nonredundant quotient flow, show that the appropriate quotient beta vanishes to the required order or exactly, and verify the target's stress-tensor and operator premises.

13 Effective envelopes and threshold matching

13.1 Different envelopes, one controlled handoff

An EFT is specified by active field content, operator basis, power counting, truncation, scale window, and error estimate. Above and below a heavy threshold, the UV and IR theories are not one unchanged Lagrangian evaluated at different μ . They are different envelopes related by matching.

Let

$$\mathcal{M} : (\mathcal{O}_{\text{UV}}, C^{\text{UV}}, \mathcal{W}_{\text{UV}}) \longrightarrow (\mathcal{O}_{\text{IR}}, C^{\text{IR}}, \mathcal{W}_{\text{IR}})$$

be the matching morphism at μ_m . It is fixed by requiring equality of a chosen set of low-energy amplitudes or Green functions through the declared order. Decoupling states that heavy effects at energies well below M are absorbed into renormalized low-energy parameters or power-suppressed local operators, subject to its assumptions [40, 41, 42].

Theorem 13.1 (EFT envelope and threshold-gluing theorem (SYM10-Q08)). *Standing-preserving continuation across a threshold requires:*

1. a UV and IR envelope with declared domains and truncation orders;
2. overlap observables or amplitudes;
3. an operator-basis and Wilson-coefficient matching map;
4. matching-scale and scheme covariance to the asserted order;
5. inherited anomaly, symmetry, and global-target data;
6. controlled truncation and threshold errors.

Dropping a heavy field, choosing a matching scale, or writing two Lagrangians with similar light fields does not by itself establish physical continuation.

Proof. The decoupling theorem supplies locality of the low-energy heavy-field effects under its assumptions. Matching fixes the finite relation between UV and IR coefficients by equating overlap observables. RG evolution above and below the threshold transports each envelope, while the matching matrix glues them at μ_m . Dependence on the arbitrary matching scale cancels in physical observables through the computed order when running and matching are combined. Anomalies and exact symmetries cannot be dropped with a heavy field; they must be matched, for example by Wess–Zumino terms or other low-energy structures when required. The listed data are therefore necessary for a lawful handoff. $\square$

13.2 Tree-level heavy-scalar example

Consider a light scalar ϕ and heavy scalar H with

$$\mathcal{L}_{\text{UV}} = \frac{1}{2}(\partial\phi)^2 - \frac{1}{2}m^2\phi^2 + \frac{1}{2}(\partial H)^2 - \frac{1}{2}M^2H^2 - \frac{\kappa}{2}H\phi^2.$$

For external momenta much smaller than M , the heavy-field equation gives

$$H = -\frac{\kappa}{2M^2}\phi^2 + \frac{\kappa}{2M^4}\square\phi^2 + \mathcal{O}(M^{-6}).$$

Substitution yields

$$\Delta\mathcal{L}_{\text{IR}} = \frac{\kappa^2}{8M^2}\phi^4 - \frac{\kappa^2}{8M^4}\phi^2\square\phi^2 + \mathcal{O}(M^{-6}),$$

up to integrations by parts and basis choices. The coefficients are matching data. A field redefinition can move the derivative operator into other basis elements without changing the on-shell amplitudes.

13.3 Boundary-trace interpretation

The AASC boundary-trace language classifies matching as a role-anchor interface between envelopes [21, 20, 19]. It does not select a preferred μ_m or prove the coefficient. The physical coefficient is fixed by the matching calculation or empirical boundary data, not by the structural label.

Adversarial check

A heavy threshold can fail to decouple when its mass is tied to a coupling that grows with the mass, when anomaly matching requires a remnant, or when the low-energy target changes. The theorem is therefore conditional on the declared decoupling and matching assumptions.

14 Dimensional transmutation and RG-invariant scale relations

14.1 From a running coordinate to an invariant relation

A classically scale-free theory can generate an RG-invariant scale when a dimensionless coupling runs. For one coupling, define

$$\Lambda_{\text{RG}} = \mu \exp \left[- \int^{g(\mu)} \frac{dg'}{\beta(g')} \right], \quad (13)$$

where the lower limit and finite convention are part of the definition. Along the RG trajectory,

$$\frac{d \log \Lambda_{\text{RG}}}{d \log \mu} = 1 - \frac{1}{\beta(g)} \frac{dg}{d \log \mu} = 0.$$

Thus Λ_{RG} is constant along the flow.

For an asymptotically free one-loop beta function

$$\beta(g) = -b_0 g^3 + \mathcal{O}(g^5), \quad b_0 > 0,$$

one obtains

$$\Lambda_{\text{RG}} = \mu \exp \left[- \frac{1}{2b_0 g^2(\mu)} \right] (1 + \text{higher-order corrections}).$$

For an $\text{SU}(N_c)$ gauge theory with N_f Dirac fermions in the fundamental representation,

$$b_0 = \frac{11N_c - 2N_f}{48\pi^2}$$

in the convention $\beta(g) = -b_0 g^3 + \dots$ [37, 38].

Theorem 14.1 (Dimensional-transmutation theorem (SYM10-Q09)). *Under the declared beta-function, scheme, and domain assumptions, (13) is invariant along the RG trajectory. Dimensional transmutation converts a dimensionless running coordinate together with an integration constant, boundary condition, matching datum, or observable input into an RG-invariant scale relation. It does not determine the numerical value of that scale without the additional datum.*

Proof. The RG invariance follows from the differential calculation above. A scheme change $g' = f(g)$ changes the finite convention of the integral and can rescale Λ_{RG} by a finite constant, while physical dimensionless ratios and observables are preserved when matching is transported. Specifying one measured coupling at one scale, one bound-state mass, or an equivalent boundary condition fixes the integration constant. Without such input, the RG equation defines a one-parameter family of trajectories and does not select a numerical member. $\square$

14.2 Role conversion, not creation ex nihilo

The classical Lagrangian may contain no mass parameter, yet the quantum theory acquires an invariant scale relation. This is role conversion: the running coupling and its boundary datum are repackaged into an RG-invariant scale. It is not an inference that the formalism predicts the observed value without input. In QCD, hadron masses are proportional to Λ_{QCD} up to quark-mass and nonperturbative effects, but the numerical value is fixed by experiment or a UV boundary condition.

14.3 Coleman–Weinberg qualification

Radiative symmetry breaking can also convert dimensionless couplings and a renormalization condition into a vacuum scale [39]. The location of a minimum in a truncated effective potential is gauge- and scheme-sensitive unless handled with the appropriate identities and observable reconstruction. The

resulting physical pole or scattering data can be meaningful; the arbitrary subtraction scale is not thereby promoted into the physical vev.

Exact nonclaim boundary

RG invariance of Λ_{RG} does not imply numerical prediction, absolute normalization, or a solution of the hierarchy problem. It also does not make every transmutation scale scheme-independent as a bare number; only properly matched physical relations are invariant.

15 IR symmetry and emergence ancestry

15.1 IR physicality is not UV provenance

An infrared symmetry can be exact at an IR fixed point, approximate over a finite scale window, or organized as an EFT expansion. It acts on physical states or operators in that target and can produce multiplets, selection rules, Ward identities, or anomaly constraints. Its physical standing does not require a known ultraviolet completion. A stronger statement—that the symmetry *emerges from a specified UV theory*—does require a transport and matching argument connecting the two targets.

Definition 15.1 (IR-symmetry packet). An IR-symmetry packet is

$$\mathfrak{S}_{\text{IR}} = (G_{\text{IR}}, \rho_{\text{IR}}, \mathcal{O}_{\text{IR}}, I_{\text{IR}}, \delta\mathcal{L}_{\text{break}}, \mathcal{A}_{\text{IR}}, \mathfrak{B}_{\text{IR}}),$$

where the entries specify the acting object, its action, operator locus, scale window, controlled symmetry-breaking deformations, anomaly data, and state or phase.

Definition 15.2 (Emergence-ancestry packet). For a claim that $\mathfrak{S}_{\text{IR}}$ emerges from a specified ultraviolet target D_{UV} , define

$$\mathfrak{E}_{\text{UV} \rightarrow \text{IR}} = (D_{\text{UV}}, \mathfrak{S}_{\text{IR}}, \mathfrak{M}_{\text{UV} \rightarrow \text{IR}}, \mathcal{A}_{\text{match}}, \Delta_{\text{UV} \rightarrow \text{IR}}),$$

where $\mathfrak{M}_{\text{UV} \rightarrow \text{IR}}$ is the RG/EFT matching map, $\mathcal{A}_{\text{match}}$ records anomaly and robust-charge transport, and $\Delta_{\text{UV} \rightarrow \text{IR}}$ records the relevant and irrelevant deformations controlling the approach to the IR target.

Theorem 15.3 (IR-symmetry and emergence-ancestry theorem (SYM10-Q10)). *An IR symmetry has physical standing when its IR target fixes the scale window or fixed point, operator locus, action on the physical quotient, state or phase, controlled breaking, and applicable anomaly data. Under those conditions it is a physical/effective action regardless of whether a UV completion is specified. If one additionally claims that this symmetry emerges from a particular UV theory, then a UV-to-IR transport or matching map, deformation ancestry, and anomaly or robust-charge matching are required. Neither clause promotes the IR symmetry to a UV primitive by default, and neither is a presentation change.*

Proof. The action on IR operators or states and its Ward identities establish quotient-nontrivial physical content directly in the IR theory. No UV map is required for that statement. An emergence claim relative to a specified UV target is stronger: the matching map identifies which UV observables and operators flow to the IR structures, while anomaly or robust-charge data and controlled deformations constrain the transport. Because these are logically additional premises, IR physicality and UV provenance must remain distinct. The nontrivial IR action also prevents either from being reduced to skin. $\square$

15.2 Fixed-point emergence

At the Wilson–Fisher point, scale and conformal symmetry emerge after the mass is tuned to the critical surface. Lattice rotational anisotropies and higher operators can become irrelevant, producing a larger long-distance spacetime and internal symmetry than a microscopic regulator displays. The exact emergence statement is target-indexed: it concerns the critical continuum limit, the operator category in which the irrelevant deformations vanish, and the RG map that connects the specified microscopic realization to that limit.

15.3 Heavy-quark symmetry

For a heavy quark with velocity v , the field decomposition

$$Q(x) = e^{-im_Q v \cdot x} (h_v(x) + H_v(x))$$

leads at leading order to the HQET Lagrangian

$$\mathcal{L}_{\text{HQET}} = \bar{h}_v i v \cdot D h_v + \mathcal{O}(1/m_Q).$$

The leading interaction is independent of the heavy-quark spin and, for multiple heavy flavors, approximately independent of the heavy mass, yielding heavy-quark spin/flavor symmetry. Chromomagnetic and kinetic operators break the symmetry at order $1/m_Q$ [56, 57]. The IR/HQET symmetry is therefore a controlled physical EFT action. Its derivation from full QCD additionally supplies an explicit emergence-ancestry packet; neither fact makes the symmetry a primitive exact symmetry of full QCD at finite mass.

15.4 Anomaly inheritance

A standalone IR symmetry carries whatever anomaly data are intrinsic to its IR target. When the symmetry is claimed to emerge from a specified UV theory, 't Hooft anomalies and other robust data must be matched or otherwise accounted for along the UV-to-IR map. Conversely, an accidental IR symmetry is possible only if the IR operator content permits it and no anomaly obstruction forbids the claimed realization. SYM-3's anomaly fields remain unchanged unless the target update changes the relevant background or operator category [8].

Adversarial check

The phrase “more symmetric in the infrared” is incomplete as an emergence claim unless the UV and IR targets, transformed operators, exact or approximate status, breaking deformations, matching map, and anomaly transport are specified. But failure to supply a UV completion does not by itself defeat the physical symmetry of a self-contained IR QFT.

16 Scale and normalization no-promotion

16.1 Scale anchors

Definition 16.1 (Scale anchor). A physical scale anchor is a dimensionful datum fixed by an independent standing-bearing witness, boundary condition, matching relation, or observable appropriate to the target. A coordinate that merely carries units or labels a subtraction procedure is not an anchor.

The support corpus separates scale from normalization and transport from fixation. A parameter can be transport-neutral without being absolutely fixed; a stable ratio can be physical without selecting the denominator; and a matching scale can be a calculational coordinate even when a nearby particle mass is a physical threshold [25, 24, 26].

Theorem 16.2 (Scale/no-promotion theorem (SYM10-Q11)). *None of the following independently fixes a standing-bearing physical scale:*

1. the renormalization scale μ or subtraction point;
2. a regulator or cutoff;
3. an arbitrary unit or normalization convention;
4. a fitted matching scale used as input;
5. naturalness, simplicity, minimality, or a solver extremum;
6. a bare mass coordinate or unanchored vev coordinate;

7. *a stable dimensionless ratio without a separate scale certificate.*

A physical scale is fixed only by an independent witness, boundary datum, matching relation, pole, response, or equivalent target-specific certificate.

Proof. A change of μ is compensated by RG transport and leaves physical predictions invariant to the computed order. Regulators and subtraction schemes are presentation data unless identified with a physical UV structure by an additional theorem. Units rescale dimensionful numbers without changing dimensionless observables. A fitted threshold or measured vev is a boundary input; its physicality does not make it internally derived. Naturalness and minimality are comparative criteria, not observables or boundary conditions. Bare parameters and cutoff contributions are scheme-dependent coordinates. A ratio determines only a relative scale unless one factor is anchored. These cases exhaust the listed selectors and each lacks independent scale authority. $\square$

16.2 What the theorem does not demote

The theorem does not say that pole masses, bound-state scales, transition temperatures, decay constants, or empirically matched thresholds are conventional. Those can be physical witnesses. It says that the displayed coordinate or preference must not be confused with the witness that fixes it.

16.3 Hierarchy comparisons

A small ratio such as v/M_{Pl} is a physical diagnostic once both scales are independently defined. The ratio does not itself determine either scale or select a UV completion. A cutoff-sensitive correction can indicate a matching sensitivity or a naturalness concern, but the cutoff term is not an observable mass contribution until a UV target and matching relation are specified. These distinctions preserve the hierarchy question while blocking scale laundering [26].

Physics translation

The correct AASC role of a physical measured scale can be anchor or boundary input. The no-promotion theorem is not a denial of physical scales; it is a prohibition on manufacturing them from presentation coordinates or diagnostic preferences.

IV

Realized vacuum-sector correspondence

17 A realized-QFT sector: homogeneous vacuum energy and finite renormalization

The general scale/RG analysis becomes especially sharp in a sector where the presentation quotient can be written explicitly. The relevant object is not the full renormalized stress tensor on arbitrary curved backgrounds. It is the homogeneous metric-proportional scalar sector used in the supplied zero-point-energy theorem. The restriction is deliberate and load-bearing.

17.1 The restricted sector

Definition 17.1 (Homogeneous metric-proportional sector). Fix a background class, matter theory, quantum state, and independently renormalized non-vacuum gravitational couplings. The sector V_0

consists of scalar assignments ρ_R entering the homogeneous matter expectation value through

$$\langle T_{\mu\nu} \rangle_{\text{hom}} = -\rho_R g_{\mu\nu}, \quad (14)$$

with the finite local metric-proportional counterterm freedom retained.

A finite renormalization may shift the coefficient by

$$\rho_R \mapsto \rho_R + \delta\rho, \quad \delta\rho \in \mathbb{R}. \quad (15)$$

Let $\mathcal{R}_{\text{fin}}$ be the equivalence relation generated by these common additive shifts. The quotient $V_0/\mathcal{R}_{\text{fin}}$ is the prediction space relevant to a scalar depending only on the absolute homogeneous matter-sector offset. This formulation follows the physics-native ZPE analysis rather than beginning from an AASC role label [13].

Theorem 17.2 (Homogeneous-vacuum quotient theorem (SYM10-Q12)). *Let $X : V_0 \rightarrow \mathbb{R}$ be a scalar proposed as a standalone prediction of the absolute homogeneous matter-sector coefficient. If X descends to $V_0/\mathcal{R}_{\text{fin}}$, then X is constant on every translation orbit. Because the action (15) is transitive on the coefficient line, no nonconstant scalar depending only on ρ_R descends to the quotient.*

Proof. Descent requires

$$X(\rho_R + \delta\rho) = X(\rho_R)$$

for every finite shift $\delta\rho$. Given any two representatives ρ_1 and ρ_2 , choose $\delta\rho = \rho_2 - \rho_1$. Then $X(\rho_2) = X(\rho_1)$. Hence X is constant on the entire orbit and cannot encode a nontrivial absolute prediction of ρ_R . $\square$

The theorem does not say that every vacuum response is zero or conventional. It says that the absolute coefficient of one fixed homogeneous sector is not a standalone renormalization-invariant scalar prediction. The distinction is the same one used throughout the paper: a representative coordinate can be indispensable in a calculation without being the invariant output of that calculation.

17.2 Relational and response quantities

Consider two physical configurations, boundary conditions, phases, or states whose renormalization convention is shifted coherently. A difference

$$\Delta\rho = \rho_R^{(1)} - \rho_R^{(2)}$$

is invariant under a common additive shift. Derivatives with respect to a physical separation, external source, temperature, order parameter, or geometric boundary datum can likewise remove the common constant. Casimir forces, phase-transition latent energies, stress differences, and response functions therefore occupy a different quotient class from the absolute offset [58, 13].

Physics translation

The theorem separates three questions that are often conflated:

1. whether a vacuum contribution enters the renormalized effective action;
2. whether an absolute matter-sector split is independently identifiable;
3. whether a relational or total gravitational response is measurable.

The first and third can be true while the second is false.

17.3 Semiclassical gravity and the split degeneracy

Write the metric-proportional part of the semiclassical field equation using a renormalized gravitational coefficient Λ_R :

$$\Lambda_{\text{eff}} = \Lambda_R + 8\pi G_R \rho_R. \quad (16)$$

The compensating transformation

$$\rho_R \mapsto \rho_R + \delta\rho, \quad \Lambda_R \mapsto \Lambda_R - 8\pi G_R \delta\rho \quad (17)$$

preserves Λ_{eff} . Thus a curvature measurement can identify the total matched coefficient, subject to the gravitational target and state, without independently identifying the split between a matter-sector offset and the gravitational counterterm. This is a parameter-identifiability statement, not a claim that the total coefficient is unphysical.

Theorem 17.3 (Semiclassical split non-identifiability theorem (SYM10-Q13)). *Within V_0 :*

1. ρ_R is a presentation-dependent homogeneous offset;
2. common-shift-invariant differences and responses are relational physical data;
3. Λ_{eff} is the matched total coefficient appearing in the semiclassical gravitational equation;
4. a numerical value of Λ_{eff} remains an empirical, boundary, or theory-input question not answered by the quotient theorem.

Proof. The first statement follows from theorem 17.2. The second follows because common additive shifts cancel. The third follows from invariance of (16) under (17). The fourth follows because invariance of the combination does not provide a numerical boundary condition for it. $\square$

17.4 What the theorem removes and what it leaves

The result invalidates the direct inference

$$\text{cutoff zero-point sum} \implies \text{standalone prediction of observed curvature.}$$

It does not solve radiative stability, naturalness, state selection, the observed smallness of Λ_{eff} , or the full renormalization problem of the stress tensor in curved spacetime. Those remain distinct scientific burdens [59, 60, 49, 50].

Exact nonclaim boundary

No statement in this section applies automatically to nonhomogeneous stress-tensor components, curvature-dependent local counterterms, boundary-dependent energies, anomaly-induced terms, dynamical gravity beyond the stated semiclassical target, or scalar quantities that depend on additional physical witnesses. Any extension beyond V_0 is a target update requiring a new theorem.

18 Standing and renormalization-quotient descent in the vacuum sector

The preceding section was stated entirely in ordinary renormalized-QFT language. This section introduces the AASC bridge only after the QFT quotient has been fixed. The purpose is not to redescribe the same result ornamentally. It is to test whether the structural standing criterion and the standard renormalization-invariance criterion select the same scalar content in a realized sector.

18.1 Binding map

Let $\mathcal{X}(V_0)$ be the declared scalar assignments on the homogeneous sector. The supplied bridge fixes the following correspondence [14, 15]:

AASC term	Physics-bound realization in V_0
admissible redescription	finite metric-proportional renormalization shift
standing-equivalence class	orbit in $V_0/\mathcal{R}_{\text{fin}}$
witness	invariant scalar, response, difference, or matched total coefficient
skin	absolute representative offset ρ_R
relational occupant	common-shift-invariant difference or response
boundary / matched input	numerical value of the total Λ_{eff} coefficient

Definition 18.1 (Physics-bound standing in V_0). A scalar assignment $X \in \mathcal{X}(V_0)$ has standing if its value is unchanged under every admitted finite-renormalization redescription and is attached to the declared physical witness rather than to a privileged representative of the orbit.

This definition is intentionally physics-bound. It does not import the complete AASC kernel into the QFT theorem. It identifies the relevant structural predicate with the already established redescription action on this target.

Theorem 18.2 (Realized standing/quotient correspondence (SYM10-A04)). For scalar assignments $X \in \mathcal{X}(V_0)$ satisfying the target and binding conditions above,

$$\text{Standing}_{V_0}(X) \iff \text{Descends}_{V_0/\mathcal{R}_{\text{fin}}}(X). \quad (18)$$

The equivalence is restricted to the homogeneous metric-proportional scalar sector and the finite-renormalization relation of [theorem 17.1](#).

Proof. Suppose X has standing. By [theorem 18.1](#), every admitted finite renormalization preserves its value, so X is constant on each $\mathcal{R}_{\text{fin}}$ orbit. Therefore a unique scalar $\tilde{X}$ exists on the quotient such that $X = \tilde{X} \circ q$, where $q : V_0 \rightarrow V_0/\mathcal{R}_{\text{fin}}$ is the quotient map. Hence X descends.

Conversely, suppose X descends. Then $X = \tilde{X} \circ q$ for some quotient scalar $\tilde{X}$ and is invariant under every finite shift generating $\mathcal{R}_{\text{fin}}$. The binding proposition of the supplied bridge identifies those shifts with the complete admitted redescription class for the declared scalar sector and identifies quotient invariance with preservation of the physical witness. Thus X satisfies the physics-bound standing predicate. $\square$

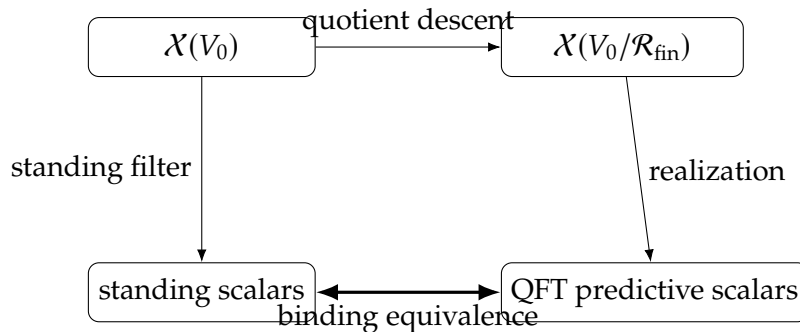


Figure 2: The standing and finite-renormalization filters coincide only after the physics-bound target and binding map are fixed.

18.2 Consequences

The equivalence gives a realized interpretation of the general quotient doctrine used earlier in the paper. A parameter can move under a lawful redescription while its physical quotient class remains fixed. In V_0 , the statement is exact enough to support a bidirectional theorem rather than a one-way analogy.

Three consequences follow immediately.

1. A nonconstant scalar of ρ_R alone has neither quotient descent nor standing.
2. A relational difference or response invariant under the common shift can have both.
3. The matched total Λ_{eff} can have standing at the coupled semiclassical target, while its decomposition into Λ_R and ρ_R remains nonidentifiable.

Adversarial check

A scalar can fail the theorem in two ways: the finite-renormalization relation may be incomplete for its target, or the AASC binding may not identify the same admissible redescrptions and witnesses. Either failure blocks extension of [theorem 18.2](#); neither is repaired by citing the theorem outside V_0 .

V

Structural integration, normal form, and handoff

19 AASC role map after the QFT classification

The physical results are now sufficiently typed to assign structural roles. The role map is deliberately many-to-many across different logical levels. A fixed point can be a target condition, a physical symmetry locus, and a source of scaling witnesses. A beta function can be a transport vector field and a trace-anomaly coefficient. A scheme parameter can be skin while the RG trajectory it coordinates is physically predictive.

Theorem 19.1 (SYM10-A01: scale and parameter role theorem). *At the frozen target, the scale-sector objects admit the following role assignment:*

1. *target-defining masses, boundary values, fixed phases, and independently witnessed scales are anchors or boundary inputs at the loci where varying them changes the physical problem;*
2. *RG trajectories, matching maps, and fixed-point manifolds are transport or structural data;*
3. *universal exponents, trace matrix elements, anomaly coefficients, and matched observables are derived witnesses;*
4. *schemes, subtraction points, field normalizations, redundant operators, and untransported coordinate choices are presentation data;*
5. *Wilson coefficients and approximate symmetry parameters are effective-envelope data;*
6. *a target-indexed IR symmetry is a physical/effective action, not skin; when an emergence-from-UV claim is made, its matching and deformation ancestry are additional transport/provenance data rather than part of the symmetry's intrinsic IR physicality.*

No role follows from numerical display alone.

Proof. The assignments follow from the quotient, variation, matching, and target tests already established. Removing a lawful presentation coordinate while retaining the reconstruction map leaves the physical quotient unchanged. Removing an anomaly coefficient, critical exponent, or overlap observable changes a derived diagnostic. Changing a boundary condition or measured scale changes the target. An IR

symmetry acts on the IR quotient and therefore is physical there. A separate UV-to-IR ancestry certificate is required only when the manuscript claims emergence from a specified UV target; its target dependence blocks primitive promotion in either case. These distinctions instantiate the imported role and mobility classifications [16, 17, 18]. $\square$

Theorem 19.2 (SYM10-A02: trace, fixed-point, and emergence role theorem). *The trace identity, fixed-point data, and emergent-symmetry record occupy distinct structural roles:*

1. $[T^\mu_\mu]_R$ and its Ward identities are diagnostic/operator structures;
2. the nonredundant or quotient beta term is a quantum scale-breaking witness and transport coefficient, while raw flavor/redundant beta motion is separately typed and is not an independent primitive scale;
3. a quotient-stable fixed point is a zero of the induced physical RG vector field and is a target-bearing transport endpoint or invariant locus;
4. a conformal symmetry, when its premise ledger is satisfied, is a physical spacetime action at that target;
5. an IR symmetry is an effective physical action at its target; a claim that it emerges from a specified UV target additionally carries matching, anomaly-transport, and deformation ancestry.

The terms cannot be collapsed into a single tag “scale symmetry.”

Proof. The trace operator records the response to local or global scaling and is not itself an action on states. The beta vector organizes RG transport and appears in the trace identity, but its raw coordinate representation can include redundant or flavor rotations. A fixed point is defined by the quotient-stable vanishing of the induced physical vector $\bar{\beta}$; conformal action requires additional current and stress-tensor conditions. IR symmetry is established by its nontrivial action and controlled violations, while emergence from a specified UV theory is a further transport-provenance statement. The roles therefore apply at different loci and remain compatible without identity. $\square$

Theorem 19.3 (EFT and matching role theorem (SYM10-A03)). *At a declared threshold, the ultraviolet and infrared descriptions are distinct effective envelopes. Their matching map is a boundary-trace continuation interface carrying overlap observables, operator-basis transport, Wilson coefficients, truncation order, and anomaly/symmetry inheritance. The matching scale is a transport or bookkeeping coordinate unless an independent physical threshold witness promotes it to target-bearing scale data.*

Proof. The physical gluing conditions are established by [theorem 13.1](#). The continuation-locus and boundary-trace imports then classify the matched handoff without replacing the QFT matching calculation [20, 21, 24]. Varying a bookkeeping matching scale while transporting coefficients and observables leaves the declared prediction invariant, whereas changing the active-field envelope, threshold witness, truncation order, or matching data changes the effective target. Hence the envelope, matching morphism, and scale coordinate occupy distinct roles. $\square$

19.1 Representative, invariant, and target-bearing objects

Object	Role	What is preserved or blocked
$g^i(\mu)$	transport coordinate	meaningful with beta vector, boundary data, and observables; not free same-domain reassignment
μ	presentation and transport coordinate	organizes logarithms; does not select a mass
raw $\beta^i \partial_i$	coordinate transport vector	may contain vertical flavor or redundant motion; not a physical fixed-point criterion by itself
$\bar{\beta}$ or $\mathcal{B}$	quotient transport vector	its vanishing defines the physical fixed locus; preserved under lawful quotient transport
$[T^\mu_\mu]_R$	diagnostic operator	trace ancestry remains typed; no generic “one anomaly” label
critical exponents	quotient witnesses	invariant after redundancy quotient and basis transport
matching map	boundary and continuation interface	connects different envelopes; matching scale not automatically physical threshold
Λ_{RG}	transmutation relation	trajectory invariant once boundary data are fixed; not internally numerically selected
G_{IR}	target-indexed physical or effective action	acts on IR states/operators without requiring a known UV completion
UV $\rightarrow$ IR ancestry	transport/provenance relation	required only when emergence from a specified UV target is claimed
ρ_R in V_0	presentation-dependent offset	no standalone absolute scalar prediction
Λ_{eff}	matched total coefficient	physical at coupled target; split remains nonidentifiable

Corollary 19.4 (Realized-QFT bridge role). *The correspondence of [theorem 18.2](#) is an exact realized-QFT witness of the structural quotient method at V_0 . It licenses the claim that standing and standard renormalization invariance coincide for the declared scalar sector. It does not license a universal identification of AASC standing with every renormalization quotient in QFT.*

Proof. The forward and reverse implications are established only because the binding map identifies the complete admitted redescription and witness data in V_0 . Outside that sector, the renormalization group, counterterm freedom, operator algebra, or witness class can be larger or differently typed. The theorem therefore supplies a realized example and a restricted equivalence, not an unrestricted schema substitution. $\square$

20 The renormalization-covariant scale normal form

20.1 Packet definition

For a fixed target D_{10} , define the scale/RG packet

$$\mathfrak{R}_{10}(D_{10}) = (\mathcal{D}_{\text{cl}}, \Theta_R, \bar{\mathcal{R}}, \mathcal{U}, \mathcal{F}_*, \mathcal{E}, \mathcal{M}, \Lambda_{\text{RG}}, \mathcal{E}_{\text{IR}}, \mathcal{Z}_0). \quad (19)$$

The entries respectively contain the classical dilation record, renormalized trace packet, quotient RG transport, universality and fixed-point data, effective envelopes, threshold matching, dimensional-transmutation relation, the IR-symmetry record together with any optional UV-emergence ancestry, and

the optional V_0 correspondence. A component is absent rather than silently set to a default when its target conditions are not met.

20.2 Compatibility interfaces

The packet is coherent only when the following interfaces hold.

1. The classical current and quantum trace use compatible stress-tensor, improvement, boundary, and operator conventions.
2. The trace beta data and anomalous dimensions generate $\overline{\mathcal{R}}$ after projection to the nonredundant physical quotient; raw flavor or redundant beta motion is retained only in its proper current/presentation locus.
3. Scheme transformations transport couplings, operators, coefficients, sources, matching maps, and observables as one coupled datum.
4. Universality and fixed-point claims use the nonredundant quotient tangent space and the declared operator category.
5. EFT envelopes are connected only by matching maps with overlap observables, error order, and anomaly/symmetry inheritance.
6. Λ_{RG} is attached to a trajectory and boundary datum rather than treated as an independently selected number.
7. IR symmetry is attached to its IR target and controlled deformations. A UV matching record is required only when an emergence-from-a-specified-UV claim is included in the packet.
8. The ZPE component is included only when the target contains the sector V_0 and the finite-renormalization binding of [theorems 17.2, 17.3 and 18.2](#).

Theorem 20.1 (Scale-sector normal-form theorem (SYM10-A05)). *For a fixed D_{10} satisfying the theorem premises of F01–F02, Q01–Q13, and A01–A04, the packet (19) is preserved under lawful presentation changes. A change in a representative coupling, scheme, field basis, operator basis, subtraction point, or redundant direction does not change the packet when all dependent data are transported coherently. A change in an invariant component or target field changes the physical scale problem or moves to a declared target fiber.*

Proof. F01 separates the acting loci and target updates. F02 assigns every parameter role and mobility before comparison. [theorem 9.1](#) and Q04 establish coherent covariance of the full renormalized datum. Q03 gives the quotient transport and composition law. Q05–Q07 identify the universality data, quotient-fixed loci, and linearized response on the nonredundant physical quotient. Q08 supplies the only lawful bridge between different EFT envelopes. Q09 distinguishes transmutation invariance from numerical selection. Q10 separates standalone IR physicality from the additional ancestry needed for an emergence-from-UV claim. Q11 blocks unlicensed scale promotion. Q12–Q13 supply the vacuum quotient and split result, while A04 supplies the optional realized standing correspondence. Each packet component is therefore either invariant under the lawful presentation map or transported to an equivalent representative; changing an untransported invariant or target field breaks one of the compatibility interfaces. $\square$

Theorem 20.2 (SYM10-MAIN: renormalization-covariant scale and emergent-symmetry normal form). *Fix a renormalized-QFT or EFT target D_{10} with a declared operator basis, physical quotient, scale interval, lawful presentation class, effective envelopes, matching maps, background and state data, symmetry/action record, and fixed-point, anomaly, or limit premises. Then:*

1. *active dilatation, RG transport, presentation change, EFT transition, and state/target change are distinct typed loci;*
2. *renormalization-scale dependence between scale-indexed descriptions is carried by a quotient-level RG transport class with identity and composition; physical dilatation scaling remains distinct;*

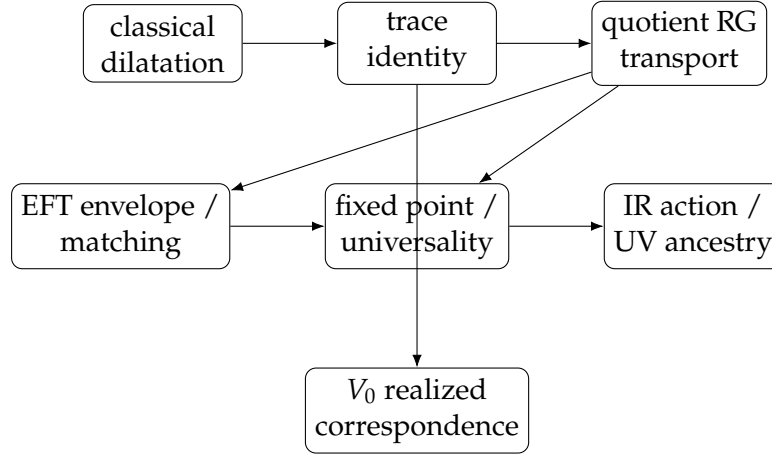


Figure 3: The normal form composes distinct scale loci without identifying them. Arrows represent theorem interfaces, not equality of the nodes.

3. *raw running parameters are coordinates on that transport and not independently reassignable same-target occupants;*
4. *scheme, regulator, normalization, field, subtraction, and redundant-operator changes are presentational only when the complete dependent datum is transported;*
5. *the trace decomposes into separately typed explicit, nonredundant beta-function, anomalous-dimension/mixing, virial/improvement, EOM/contact, and background terms, with raw flavor/redundant running kept in its proper current or presentation sector;*
6. *fixed points are zeros of the induced quotient RG vector field and universality classes are quotient-stable structures; raw beta components may remain along redundant or flavor directions, while conformal enhancement remains conditional on its premise ledger;*
7. *relevant, irrelevant, marginal, and redundant directions are classified on the nonredundant quotient tangent space;*
8. *EFT thresholds are matched interfaces between different envelopes rather than silent continuation of one description;*
9. *dimensional transmutation is an RG-invariant role conversion requiring boundary, matching, or observable input for numerical scale fixation;*
10. *an IR symmetry is a target-indexed physical/effective action with controlled breaking and anomaly data; when emergence from a specified UV theory is claimed, UV-to-IR matching, deformation ancestry, and anomaly transport are additional required data;*
11. *no arbitrary scale, cutoff, subtraction point, fitted threshold, heuristic, bare coordinate, or unanchored $\vev$ becomes a physical scale selector without an independent witness;*
12. *when the target includes V_0 , the absolute homogeneous offset fails quotient descent, the semiclassical split is non-identifiable, and scalar AASC standing is equivalent to descent to the ordinary finite-renormalization quotient.*

Consequently, the scale-sector physical content is represented by $\mathfrak{R}_{10}(D_{10})$ up to lawful presentation, with target changes and invariant differences kept explicit.

Proof. Items 1–4 are F01–F02, the normalization lock, and Q03–Q04. Item 5 is Q02, anchored classically by Q01. Items 6–7 are Q05–Q07. Item 8 is Q08. Item 9 is Q09. Item 10 is Q10. Item 11 is Q11. Item 12 is Q12–Q13 together with A04. Their compatibility and presentation covariance are established by A05, while A01–A03 provide the role translation only after the QFT results are fixed. Thus the packet gives a well-typed normal form at the declared target. No clause asserts numerical parameter closure, universal

theorem 20.2 does not derive Standard Model beta functions from AASC, predict the observed couplings or masses, prove that the observed Standard Model is scale or conformally invariant, solve the hierarchy problem, determine Λ_{RG} numerically without input, solve cosmological-constant naturalness, or extend the V_0 correspondence to arbitrary renormalized observables. It is a target-indexed normal-form theorem, not an atlas-wide exhaustion result.

status, scheme class, scale interval, effective envelope, matching data when applicable, state or phase, and role/mobility profile exported by SYM-10. A capstone row that collapses active symmetry, RG transport, scheme redescription, EFT emergence, and scale anchoring is ill typed. Candidate completeness and same-scope exhaustion remain SYM-11 burdens.

Proof. F01 exhibits the nonidentical loci. Q02, Q03, Q08, and Q10 give different physical certificates for trace response, RG continuation, envelope transition, standalone IR action, and UV-to-IR emergence ancestry. Q11 shows that scale anchoring requires a separate witness. Since these records differ in acted-on object, quotient, exactness, anomaly, target, and role, collapsing them loses fields necessary to decide their status. The present manuscript supplies the typed rows and dependencies but not the capstone's candidate-generation proof. $\square$

21.3 Conclusion

The renormalization group does not become less physical when its coordinates are treated as coordinates. On the contrary, the quotient and transport structure reveals what the physical statement is. A running coupling belongs to a trajectory with boundary data and observable reconstruction. A raw beta function is a coordinate transport vector and can contain redundant or flavor-rotation motion; the induced quotient beta is the fixed-locus criterion and enters the physical trace statement after those vertical directions are properly separated. A fixed point is a quotient-stable zero of that induced vector, while conformal enhancement requires its own premise ledger. Universality removes microscopic distinctions only at a declared long-distance target. EFT matching connects different envelopes rather than pretending they are one unchanged description. Dimensional transmutation produces an invariant relation but does not select its numerical value without input. An IR symmetry is physically real at its own target without requiring a known UV completion. Calling it emergent from a specified UV theory is a stronger provenance claim that requires matching, deformation ancestry, and anomaly transport; neither statement makes the symmetry primitive in the UV.

The homogeneous vacuum sector adds a concrete realized bridge. There, the ordinary finite-renormalization quotient and the structural standing quotient select the same scalar predictive content. That result removes the absolute matter-sector zero-point offset from the standalone prediction space while retaining relational responses and the matched total gravitational coefficient in their proper roles. It is exact where proved and deliberately silent elsewhere.

The final picture is therefore neither “all scales are conventional” nor “every displayed scale is physical.” Scale-sector reality is the residue that survives the appropriate quotient, transport, matching, state, and witness conditions. The packet $\mathfrak{N}_{10}(D_{10})$ records that residue in a form that is covariant under lawful redescription and ready for the SYM-11 atlas without pre-empting its exhaustion theorem.

VI

Reference and reproducibility appendices

A Notation and convention dictionary

Table 1: Primary notation.

Symbol	Meaning
D_{10}	Complete renormalized-QFT/EFT target fingerprint.

Symbol	Meaning
$\mathcal{T}_R$	Renormalized theory, field content, gauge/global structure, interaction basis, quantum regime, and state/phase.
$\mathcal{O}_R$	Renormalized composite-operator basis, mixing, EOM/redundant-operator policy, sources, and improvement data.
$\mathcal{Q}_{\text{phys}}$	Physical quotient appropriate to the target.
I_μ	Scale interval or scale-indexed family.
$\mathfrak{S}$	Lawful presentation class: schemes, regulators, subtraction prescriptions, field and operator coordinates.
$\mathfrak{E}$	EFT envelope, power counting, truncation, and domain of validity.
$\mathfrak{M}$	Threshold and matching data.
$\mathfrak{B}$	Background geometry, boundary data, and external sources.
$\mathfrak{V}$	Vacuum, thermal state, phase, fixed-point target, or IR regime.
$\mathfrak{P}$	Premise bundle for scale/conformal, universality, emergence, or vacuum-correspondence claims.
$\text{Desc}_\mu(D_{10})$	Scale-indexed descriptions at μ .
$\sim_{\text{pres}}$	Lawful presentation equivalence.
$\text{Phys}_\mu(D_{10})$	Physical quotient $\text{Desc}_\mu(D_{10})/\sim_{\text{pres}}$.
$\overline{\mathcal{R}}_{\mu \rightarrow \nu}$	Descended RG transport map.
g^i	Dimensionless coupling coordinates.
m^a	Relevant mass coordinates.
C_A	Wilson coefficients.
$[O_A]$	Renormalized composite operators.
β^i	RG vector-field components.
γ_A^B	Composite-operator anomalous-dimension matrix.
M^i_j	Stability matrix $\partial_j \beta^i _*$.
θ_a	Critical exponent in the convention $Mv_a = -\theta_a v_a$.
$\mathcal{M}$	Threshold matching morphism.
Λ_{RG}	Dimensional-transmutation invariant.
$\mathcal{E}_{\text{IR}}$	Emergent-symmetry packet.
ρ_R	Renormalized homogeneous matter-sector vacuum coefficient in V_0 .
Λ_R	Renormalized geometric cosmological counterterm.
Λ_{eff}	Total coefficient $\Lambda_R + 8\pi G_R \rho_R$.
$\mathfrak{N}_{10}(D_{10})$	Integrated scale-sector packet.

A.1 Sign conventions

The metric signature is $(-, +, +, +)$ in the vacuum-sector discussion. The homogeneous vacuum stress tensor is

$$\langle T_{\mu\nu} \rangle_{0,R} = -\rho_R g_{\mu\nu}.$$

The semiclassical equation is

$$G_{\mu\nu} + \Lambda_R g_{\mu\nu} = 8\pi G_R \langle T_{\mu\nu} \rangle_R,$$

so

$$\Lambda_{\text{eff}} = \Lambda_R + 8\pi G_R \rho_R.$$

The Callan–Symanzik and anomalous-dimension signs are fixed by (4). Sources using the opposite sign convention are translated before import.

A.2 Trace-equality levels

The notation $[T^\mu_\mu]$ can denote:

1. a local renormalized operator identity;
2. an equality inside separated-point correlators;
3. an equality modulo equations of motion, BRST-exact terms, or contact terms;
4. an integrated Ward identity under boundary assumptions.

Every theorem states the level needed. No equality is silently promoted from one level to another.

B Selected trace, scheme, and quotient derivations

B.1 Local scale Ward identity

Let $W[g_{\mu\nu}, \lambda^I(x), J_A(x)]$ be the renormalized generating functional with metric, spacetime-dependent couplings, and composite-operator sources. A local Weyl variation acts schematically by

$$\Delta_\sigma = \int d^d x \, \sigma(x) \left(2g_{\mu\nu} \frac{\delta}{\delta g_{\mu\nu}} - \beta^I \frac{\delta}{\delta \lambda^I} + \cdots \right).$$

The renormalized response has the form

$$\Delta_\sigma W = \int d^d x \sqrt{|g|} (\sigma \mathcal{A} + \partial_\mu \sigma \mathcal{B}^\mu + \cdots),$$

where $\mathcal{A}$ is a local background anomaly and the omitted terms include source, flavor, and improvement contributions. Differentiation with respect to the metric and sources yields the trace identity used in [theorem 6.1](#). The identity must be interpreted at the level stated by the source: as a renormalized operator identity, in separated-point correlators, or modulo contact/EOM terms [33, 34, 32].

B.2 Why the beta vector, not its coordinate list, is geometric

Write the coupling manifold locally as $\mathcal{M}$ with vector field

$$\beta = \beta^i(g) \partial_i.$$

Under $g' = f(g)$,

$$\beta = \beta^j \frac{\partial f^i}{\partial g^j} \partial'_i = \beta'^i \partial'_i.$$

Thus the integral curves and zero locus are invariant under an invertible chart change. A coordinate coefficient can move without changing the vector field. This is the geometric content behind Q03 and Q04.

B.3 Operator coefficients as dual coordinates

Let O be a column of operators and C a row of coefficients. The interaction is CO . For $O' = MO$,

$$C' = CM^{-1}$$

is forced by

$$C'O' = CM^{-1}MO = CO.$$

A transformation of only one side is not a new scheme for the same effective action; it is a different action or an incomplete record.

B.4 Descent proof as a quotient universal property

Let $q_\mu : \text{Desc}_\mu \rightarrow \text{Phys}_\mu$ be the quotient map. If

$$d \sim_{\text{pres}} d' \implies \mathcal{R}_{\mu \rightarrow \nu}(d) \sim_{\text{pres}} \mathcal{R}_{\mu \rightarrow \nu}(d'),$$

then $q_\nu \circ \mathcal{R}_{\mu \rightarrow \nu}$ is constant on the fibers of q_μ . The quotient universal property gives a unique $\bar{\mathcal{R}}_{\mu \rightarrow \nu}$ satisfying

$$\bar{\mathcal{R}}_{\mu \rightarrow \nu} \circ q_\mu = q_\nu \circ \mathcal{R}_{\mu \rightarrow \nu}.$$

No claim of semantic completeness is needed. The theorem concerns the descriptions and equivalence relation declared by the target.

B.5 Vacuum-shift arithmetic

The coupled finite transformation is

$$(\rho_R, \Lambda_R) \mapsto (\rho_R + \delta\rho, \Lambda_R - 8\pi G_R \delta\rho).$$

Then

$$\Lambda_R - 8\pi G_R \delta\rho + 8\pi G_R (\rho_R + \delta\rho) = \Lambda_R + 8\pi G_R \rho_R.$$

The cancellation is elementary; the scientific content lies in the renormalization theorem establishing that this is the admitted finite counterterm freedom in the restricted sector.

B.6 Further explicit checks

B.7 Scheme transformation through fourth order

Let

$$\beta(g) = b_1 g^2 + b_2 g^3 + b_3 g^4 + \mathcal{O}(g^5), \quad g' = g + a g^2.$$

The inverse series is

$$g = g' - a g'^2 + 2a^2 g'^3 + \mathcal{O}(g'^4).$$

The pushforward is

$$\beta'(g') = (1 + 2a g) \beta(g).$$

Expanding first in g and then in g' yields

$$\beta'(g') = b_1 g'^2 + b_2 g'^3 + (b_3 - a b_2 - a^2 b_1) g'^4 + \mathcal{O}(g'^5).$$

This calculation is encoded in the project only as an illustrative algebra check. The scientific theorem is the coordinate-covariance argument in the main text.

B.8 Similarity of the stability matrix

For $g^i = f^i(g)$,

$$\beta^i(g') = J^i_j(g) \beta^j(g), \quad J^i_j = \frac{\partial f^i}{\partial g^j}.$$

Differentiate with respect to g^k and evaluate at a fixed point:

$$M^i_k = \frac{\partial g^\ell}{\partial g^k} \frac{\partial}{\partial g^\ell} (J^i_j \beta^j)_* = J^i_j M^j_\ell (J^{-1})^\ell_k,$$

because the term $(\partial_\ell J^i_j) \beta^j$ vanishes at $\beta(g_*) = 0$. Hence the spectrum is invariant before quotienting, and the physical spectrum is the induced spectrum on the nonredundant quotient tangent space.

B.9 Wilson–Fisher fixed point

For the $O(N)$ model in the normalization used in the text,

$$\beta(u) = -\epsilon u + \frac{N+8}{6}u^2 + \mathcal{O}(u^3).$$

The interacting zero is

$$u_* = \frac{6\epsilon}{N+8} + \mathcal{O}(\epsilon^2).$$

The mass anomalous dimension gives

$$\nu^{-1} = 2 - \frac{N+2}{N+8}\epsilon + \mathcal{O}(\epsilon^2).$$

These coordinate formulas illustrate the fixed-point theorem; the universality claim concerns the quotient-stable exponents and operator data.

B.10 Dimensional-transmutation invariant

Define

$$F(g) = \int^g \frac{dg'}{\beta(g')}.$$

Since $dg/d\log\mu = \beta(g)$,

$$\frac{dF}{d\log\mu} = 1.$$

Therefore

$$\frac{d}{d\log\mu} (\log\mu - F(g(\mu))) = 0,$$

which proves (13). The arbitrary lower limit of F is the multiplicative convention for Λ_{RG} .

B.11 Tree-level heavy-field elimination

For

$$\mathcal{L}_H = \frac{1}{2}(\partial H)^2 - \frac{1}{2}M^2 H^2 - \frac{\kappa}{2}H\phi^2,$$

the equation of motion is

$$(\square + M^2)H = -\frac{\kappa}{2}\phi^2.$$

The low-momentum expansion of $(M^2 + \square)^{-1}$ gives

$$H = -\frac{\kappa}{2M^2}\phi^2 + \frac{\kappa}{2M^4}\square\phi^2 + \cdots.$$

Substitution produces a local expansion in M^{-2} . Integration by parts and field redefinitions move terms among equivalent operator bases; on-shell amplitudes fix the matching class.

B.12 Translation quotient in V_0

Let the additive group $(\mathbb{R}, +)$ act on ρ_R by translations. The orbit of every point is all of $\mathbb{R}$, so the quotient of a single absolute coordinate is one point. A scalar invariant X satisfies

$$X(\rho) = X(\rho + C) \quad \forall C,$$

and is constant. For a family $E(a)$ with common shift, the quotient retains differences and derivatives. This is the complete invariant-theory content used in [theorem 17.2](#).

C Worked example matrix

Table 2: Representative, invariant, and role distinctions in the worked examples.

Example	Representative coordinates	Quotient-stable content	Structural status
Massless classical Yang–Mills	g in a classical action, canonical stress tensor	Classical dilatation current after improvement/boundary conditions	Physical scale action at classical target
Quantum Yang–Mills	$g(\mu)$, subtraction scheme, operator basis	Trace matrix elements, RG trajectory, matched observables, Λ_{RG} relations	RG transport plus trace witness
Wilson–Fisher fixed point	Coordinate u_* and microscopic lattice couplings	Nonredundant exponents, fixed-point operator/symmetry data	Quotient-stable fixed structure
Nonlinear scheme map	g versus $g' = g + a g^2$	Physical correlators and critical exponents	Presentation skin when complete package transported
Heavy scalar EFT	UV heavy field, matching scale, chosen operator basis	Low-energy amplitudes through truncation order	Matched envelope continuation
Heavy-quark symmetry	Velocity field h_v , HQET basis, matching scale	Spin/flavor multiplets and relations up to $1/m_Q$ corrections	Emergent/effective physical action
QCD transmutation	$g_s(\mu)$ and scheme convention	Dimensionless ratios and matched Λ_{QCD} relations	Transport-defined invariant plus boundary input
Homogeneous vacuum	ρ_R , regulator sum, finite counterterm split	Differences, responses, Λ_{eff}	Skin, relational witnesses, and matched total

C.1 Representative versus invariant checklist

For every scale-sector calculation, the following questions should be answered.

1. What is the source target and operator basis?
2. What is the physical quotient?
3. Which coordinate is running, rescaled, or redefined?
4. Which operators, Wilson coefficients, sources, and observables transform with it?
5. What is the overlap witness?
6. Is a threshold or phase boundary crossed?
7. Which data are universal, scheme-covariant, empirical, or target-dependent?
8. What exact nonclaim prevents promotion of the coordinate into an anchor?

C.2 Detailed fixed-point, transmutation, and EFT examples

C.3 Wilson–Fisher fixed point

Use the convention-normalized quartic coupling u in the $O(N)$ model near $d = 4 - \epsilon$:

$$\beta(u) = -\epsilon u + \frac{N+8}{6}u^2 + O(u^3).$$

The nontrivial zero is

$$u_* = \frac{6\epsilon}{N+8} + O(\epsilon^2).$$

The derivative

$$\beta'(u_*) = \epsilon + O(\epsilon^2)$$

shows that the quartic perturbation decays toward the infrared in the sign convention of Appendix A. The mass operator supplies the relevant direction that must be tuned to reach the critical surface. The numerical coefficient depends on the normalization of u ; the existence of the fixed point and the critical exponents, calculated consistently, are the physical statements [35, 4].

C.4 Asymptotic freedom and the transmutation relation

For a non-Abelian gauge theory,

$$\mu \frac{dg}{d\mu} = -\frac{b_0}{16\pi^2}g^3 + O(g^5), \quad b_0 = \frac{11}{3}C_A - \frac{4}{3}T_R n_f.$$

At one loop,

$$\Lambda = \mu \exp \left[-\frac{8\pi^2}{b_0 g^2(\mu)} \right]$$

is invariant along the trajectory, up to higher-loop and convention-dependent finite normalization. Its measured numerical value requires a boundary condition such as a coupling at a reference scale [37, 38].

C.5 Banks–Zaks as a controlled fixed-point example

For matter content near the loss of asymptotic freedom, the first two beta coefficients can have opposite signs and yield a weakly coupled infrared zero. This demonstrates that fixed-point existence depends on the complete matter and beta-function target, not on a generic word “gauge theory.” SYM-10 uses it as a controlled example and not as a Standard Model fixed-point claim [61].

C.6 Heavy-field decoupling and matching

Suppose a heavy field of mass M is integrated out. The low-energy action contains local operators

$$\mathcal{L}_{\text{IR}} = \mathcal{L}_{\text{light}} + \sum_{d>4} \frac{C_i^{(d)}(\mu)}{M^{d-4}} \mathcal{O}_i^{(d)}.$$

Matching at $\mu_m \sim M$ fixes the coefficients by comparing overlap Green functions or amplitudes. In a mass-independent subtraction scheme, decoupling is not automatic in the running couplings; the heavy field must be removed through an explicit matching relation. Residual μ_m dependence at finite order estimates omitted terms rather than defining a new physical threshold [40, 41, 42].

C.7 Heavy-quark symmetry as EFT emergence

For $m_Q \gg \Lambda_{\text{QCD}}$, the leading heavy-quark effective action loses sensitivity to the heavy-quark spin and, for several heavy flavors, to its mass label. The symmetry is exact only in the $m_Q \rightarrow \infty$ target and is violated by controlled $1/m_Q$ corrections. It therefore exemplifies Q10: a physically useful emergent action with a scale window and matching ancestry, not a UV primitive symmetry [56, 57].

D Atlas, target, theorem, equation, and source registries

The tables in this appendix are generated from the canonical machine-readable release records. They are publication projections of the bounded audit layer. The scientific authority remains the theorem text and its cited mathematical and physical sources; these tables do not establish semantic completeness of every possible scale-related structure.

D.1 Stable theorem registry

ID	Result	Authority	Status	Dependencies / use	Location
SYM10-F01	Scale-target and locus freeze	local typed theorem	proved	all later results	section 5
SYM10-F02	Parameter-role and mobility normal form	AASC support synthesis	proved	Q03-Q11 and A01-A03	section 5
SYM10-Q01	Classical scale-action theorem	Noether and stress-tensor proof	proved at declared scope	Q02 and A01	section 6
SYM10-Q02	Renormalized trace-identity theorem	primary QFT plus local synthesis with quotient/improved beta distinction	proved at declared scope	Q06 and A02	section 7
SYM10-Q03	RG quotient-transport theorem	Callan-Symanzik plus quotient proof	proved conditional on domain	Q04-Q10 and A05	section 9
SYM10-Q04	Scheme and presentation covariance with extended chain rule	local covariance proof on scale-independent and explicit-scale scheme maps	proved	Q05-Q10 and A05	section 10
SYM10-Q05	Universality quotient theorem	primary RG plus quotient synthesis	proved at declared target	Q06-Q07 and A02	section 11
SYM10-Q06	Quotient-fixed-point and conformal-status theorem	primary QFT with quotient-flow and premise ledger	conditional conformal conclusion	Q07 and A02	section 12
SYM10-Q07	Nonredundant quotient-response theorem	linearized induced-flow proof	proved at linearized scope	A02 and A05	section 13
SYM10-Q08	EFT envelope and threshold-gluing theorem	decoupling and matching plus boundary interface	proved conditional on matching	Q10 and A03	section 14
SYM10-Q09	Dimensional-transmutation theorem	beta-function integration	proved at declared interval	Q11 and A05	section 15
SYM10-Q10	IR-symmetry and emergence-ancestry theorem	IR QFT status plus optional UV-to-IR matching synthesis	proved at target	A02 and A05	section 16
SYM10-Q11	Scale and normalization no-promotion theorem	Q03-Q10 plus structural no-promotion	proved	A01-A03 and A05	section 17
SYM10-Q12	Homogeneous-vacuum quotient theorem	physics-native finite-renormalization proof	proved only in V0	Q13 and A04	section 18
SYM10-Q13	Semiclassical split non-identifiability theorem	compensating-shift proof	proved only in coupled V0 target	A04 and MAIN	section 18
SYM10-A01	Scale and RG role theorem	AASC composition after QFT	proved	A05 and MAIN	section 20
SYM10-A02	Trace, quotient-fixed-point, and IR/emergence role theorem	AASC composition after QFT	proved	A05 and MAIN	section 20
SYM10-A03	EFT and matching role theorem	AASC composition after Q08	proved	A05 and MAIN	section 20

ID	Result	Authority	Status	Dependencies / use	Location
SYM10-A04	Realized standing and quotient correspondence	ZPE binding theorem	proved only in V0	A05 and MAIN	section 19
SYM10-A05	Scale-sector normal-form theorem	F, Q, and A theorem composition	proved at fixed target	MAIN and SYM-11	section 21
SYM10-MAIN	Renormalization-covariant scale and emergent-symmetry normal form	integrated theorem composition	proved at declared scope	SYM-11	section 21

D.2 Frozen target family

ID	Target	Theory	Physical quotient	Scale interval	Envelope	Matching	Status
D10-CORE	Renormalized-QFT transport target	Renormalized QFT with closed operator basis	Gauge or BRST plus lawful presentation quotient	Connected scale interval	Single renormalized envelope	Explicit at boundaries	core
D10-CL	Classical dilation target	Classical local field theory	Field-coordinate and improvement equivalence	Not RG-indexed	Classical theory	Not applicable	fiber
D10-FP	Fixed-point and conformal target	Renormalized QFT at quotient-stable beta zero	Scheme and redundant-operator quotient	Neighborhood of fixed locus	Fixed theory or fixed manifold	UV or IR ancestry when claimed	fiber
D10-EFT	Matched EFT envelope family	UV and IR EFTs with distinct active fields	Field, basis, and scheme quotient in each envelope	Overlapping threshold windows	UV and IR envelopes	Explicit threshold matching map	fiber
D10-IR	Standalone IR-symmetry target	Self-contained IR QFT, EFT, or fixed-point theory	IR physical quotient	Declared IR window or fixed locus	IR EFT or fixed-point envelope	Not required for standalone IR physicality	fiber
D10-EM	UV-to-IR emergence-ancestry target	Specified UV theory plus an IR QFT, EFT, or fixed-point target	Physical quotient in each target plus lawful transport relation	Declared UV-to-IR range and IR window	UV theory or EFT plus IR EFT or fixed-point envelope	Explicit UV-to-IR RG or EFT matching map required	fiber
D10-V0	Homogeneous vacuum bridge target	Renormalized matter QFT plus semiclassical gravity split	Common additive finite-renormalization shifts	Fixed-background target	Single restricted sector	Coupled Lambda_R and rho_R shift	restricted fiber

D.3 Source-provenance ledger

ID	Source	Authority	Imported use	Bibliography key(s)	Verification
SRC-AASC-CONSTANTS	A Classification Theorem for Physical Constants	AASC import	five role classes	PhysicalConstantsClass	supplied PDF
SRC-AASC-FIXITY	The Standing Fixity of Physical Constants	AASC import	fixed-domain and continuation discipline	ConstantsFixity	supplied PDF
SRC-AASC-MOBILITY	Why Some Parameters Run and Others Cannot	AASC import	five mobility modes	ParametersRun	supplied PDF
SRC-AASC-RG	Renormalization as Standing-Preserving Scale Transport	AASC import	quotient transport, universality, EFT, and transmutation roles	RGStanding	supplied PDF
SRC-AASC-PARAMA	Parameter Admissibility and Redundancy Elimination	AASC import	redundant-parameter removal	ParameterAdmissibility	supplied PDF
SRC-AASC-PARAMB	Transport-Closure Locus for Continuous Parameters	AASC import	continuous-parameter transport classification	TransportClosure	supplied PDF
SRC-AASC-PARAMG	Scale Parameters and Cross-Envelope Matching	AASC import	matching-scale boundary status	ScaleMatching	supplied PDF
SRC-AASC-SCALE	Scale and Normalization Exactness	AASC import	scale and normalization no-promotion	ScaleNormExact	supplied PDF
SRC-AASC-CONT	Continuation-Locus Induction	AASC import	persistence through lawful change	ContinuationLocus	supplied PDF
SRC-AASC-BTRACE	Boundary-Trace Fixation	AASC import	role anchor at matching boundary	BoundaryTrace	supplied PDF
SRC-AASC-GAUGE	Gauge-Coupling Parameters and Forced Uniqueness	AASC example	scheme and normalization example	GaugeCouplingUniqueness	supplied PDF
SRC-AASC-HIER	The Hierarchy Problem and Scale Standing	AASC negative control	hierarchy no-promotion	HierarchyScale	supplied PDF
SRC-ZPE-PHYS	Why Zero-Point Vacuum Energy Is Not a Cosmological Prediction in QFT	physics-native support	finite-renormalization quotient theorem	ZPEPhysics	supplied PDF
SRC-ZPE-PROJ	Invariance Constraints and the Non-Predictivity of ZPE	AASC projection	role map of the physics theorem	ZPEAASCProjection	supplied PDF
SRC-ZPE-BRIDGE	Renormalization Invariance and Standing Preservation in ZPE	binding theorem	standing iff quotient descent in V0	ZPEEquivalence	supplied PDF
SRC-QFT-NOETHER	Noether 1918 and stress-tensor literature	primary QFT	classical dilatation current and improvement	Noether1918; CCJ1970	primary bibliography
SRC-QFT-CS	Callan and Symanzik 1970	primary QFT	Callan-Symanzik scale equations	Callan1970; Symanzik1970	primary bibliography
SRC-QFT-TRACE	Trace-anomaly literature	primary QFT	renormalized trace identity	AdlerCollinsDuncan1977; CollinsDuncan-Joglekar1977	primary bibliography
SRC-QFT-RG	Wilson, Fisher, Kogut, and Polchinski	primary QFT	universality and fixed-point RG	WilsonFisher1972; WilsonKogut1974; Polchinski1984	primary bibliography
SRC-QFT-EFT	Appelquist-Carazzone and EFT literature	primary QFT	decoupling and matching	AppelquistCarazzone1975; Georgi1993; ManoharEFT	primary bibliography
SRC-QFT-LIMIT	Fortin, Grinstein, and Stergiou - Limit Cycles and Conformal Invariance	primary QFT	raw beta versus quotient or flavor-improved RG flow on recurrent conformal trajectories	FortinGrinsteinStergiou2013	primary arXiv and JHEP bibliographic record

D.4 Dependency contracts

ID	Source	Permitted interface	Blocked inference
IMP-SYM0	SYM-0	typed symmetry record; quotient test; target updates	No scale or RG theorem
IMP-SYM2	SYM-2	active and passive spacetime distinction; stress-tensor premises	No quantum trace identity
IMP-SYM3	SYM-3	anomaly taxonomy and obstruction transport	No RG theorem
IMP-SYM5	SYM-5	basis, spurion, and invariant discipline	No scheme theorem
IMP-SYM6	SYM-6	effective and emergent symmetry interfaces	No general fixed-point theorem
IMP-SYM7	SYM-7	operator-dimension and EFT-scope indexing	No matching theorem
IMP-SYM9	SYM-9	global-form and generalized-symmetry target fields	No scale or RG classification
IMP-AASC-01	Renormalization as Standing-Preserving Scale Transport	Scale-indexed descriptions, quotient transport, universality, fixed points, EFT envelopes, matching, and transmutation roles	Does not derive beta functions or trace identities
IMP-AASC-02	Why Some Parameters Run and Others Cannot	Five mobility modes and running or nonrunning discipline	No beta-function derivation
IMP-AASC-03	A Classification Theorem for Physical Constants	Five role classes	No numerical constant prediction
IMP-AASC-04	The Standing Fixity of Physical Constants	Fixed-domain and continuation discipline	No denial of lawful running
IMP-AASC-05	Gauge-Coupling Parameters and Forced Uniqueness under Admissible Redescription	Gauge-coupling specialization and transmutation distinction	No numerical coupling selection
IMP-AASC-06	Parameter Admissibility and Redundancy Elimination - SM-PARAM-A	Redundancy elimination	No RG dynamics
IMP-AASC-07	Transport-Closure Locus for Continuous Parameters - SM-PARAM-B	Continuous-parameter transport classification	No trajectory without witnesses
IMP-AASC-08	Scale Parameters and Cross-Envelope Matching - SM-PARAM-G	Cross-envelope scale role	No absolute scale derivation
IMP-AASC-09	Scale and Normalization Exactness	Scale or normalization separation and no-promotion	No QFT beta or matching calculation
IMP-AASC-10	Boundary-Trace Fixation of Continuation Loci under AASC	Boundary-trace role anchor for matching	Not canonical representative selection
IMP-AASC-11	Continuation-Locus Induction under Admissibility-Bearing Governance	Continuation-locus identity	No dynamics
IMP-AASC-12	The Hierarchy Problem and Scale Standing	No-promotion negative control	No hierarchy solution
IMP-ZPE-01	Why Zero-Point Vacuum Energy Is Not a Cosmological Prediction in QFT	Finite-renormalization quotient, relational survivors, and split non-identifiability	Restricted to homogeneous scalar sector
IMP-ZPE-02	Invariance Constraints and the Non-Predictivity of Zero-Point Vacuum Energy in QFT	Role projection	Does not replace QFT proof
IMP-ZPE-03	Renormalization Invariance and Standing Preservation in ZPE	Standing iff quotient descent in V0	No extension to full stress tensor

D.5 SYM-10 atlas projection

Record	Object	Target	Acted upon	Quotient action	Status	Role	Theorem IDs
SYM10-ATLAS-DIL	Active dilatation	D10-CL	fields and operators	nontrivial when premises hold	classical exact or limit-dependent	physical action and domain anchor	SYM10-F01;SYM10-Q01;SYM10-A01
SYM10-ATLAS-PASSIVE	Passive scale or unit redescription	D10-CORE	coordinates and units	trivial after lawful transport	presentation-covariant	skin	SYM10-F01;SYM10-Q04
SYM10-ATLAS-TRACE	Renormalized trace identity	D10-CORE	renormalized operator algebra	diagnostic	quantum and target-dependent; raw and quotient beta distinguished	derived witness or obstruction	SYM10-Q02;SYM10-A02
SYM10-ATLAS-RG	Quotient RG transport	D10-CORE	theory-space quotient	nontrivial transport	quantum exact at declared domain	transport structure	SYM10-Q03;SYM10-A01
SYM10-ATLAS-SCHEME	Scheme and field reparameterization	D10-CORE	coupling and operator coordinates	trivial under coherent transport	presentation-dependent	skin	SYM10-Q04;SYM10-A01
SYM10-ATLAS-UNIV	Universality class	D10-FP	IR quotient data	nontrivial quotient class	fixed-point or scaling target	quotient-stable structure	SYM10-Q05;SYM10-A02
SYM10-ATLAS-FP	RG fixed point	D10-FP	nonredundant theory space	nontrivial invariant locus	quantum quotient-fixed-point status	zero of induced physical RG flow; transport endpoint or target datum	SYM10-Q06;SYM10-A02
SYM10-ATLAS-CONF	Conformal action	D10-FP	states and operators	nontrivial	conditional on premise ledger	physical spacetime action	SYM10-Q06;SYM10-A02
SYM10-ATLAS-REL	Relevant or irrelevant directions	D10-FP	quotient tangent space	nontrivial response	linearized fixed-point data	derived response structure	SYM10-Q07
SYM10-ATLAS-EFT	EFT envelope	D10-EFT	active fields and operators	target-indexed	effective and truncation-dependent	effective envelope	SYM10-Q08;SYM10-A03
SYM10-ATLAS-MATCH	Threshold matching map	D10-EFT	UV and IR predictions	nontrivial interface	conditional on overlap	boundary-trace continuation	SYM10-Q08;SYM10-A03
SYM10-ATLAS-LAMBDA	Dimensional-transmutation scale relation	D10-CORE	RG trajectory and boundary data	quotient-invariant relation	scheme-qualified	transport-defined witness	SYM10-Q09;SYM10-Q11
SYM10-ATLAS-IRSYM	IR physical symmetry	D10-IR	IR states and operators	nontrivial at IR target	effective or fixed-point exact	physical or effective action	SYM10-Q10;SYM10-A02
SYM10-ATLAS-EMERG	UV-to-IR emergence ancestry	D10-EM	UV and IR operator or observable records	nontrivial transport provenance	conditional on matching and ancestry	transport and provenance relation	SYM10-Q10;SYM10-A02
SYM10-ATLAS-MU	Renormalization scale	D10-CORE	subtraction coordinate	presentation or transport coordinate	not a physical anchor by itself	skin or transport coordinate	SYM10-Q11;SYM10-A01
SYM10-ATLAS-RHO	Homogeneous vacuum offset	D10-V0	metric-proportional scalar	quotient-trivial absolute offset	finite-renormalization dependent	skin	SYM10-Q12
SYM10-ATLAS-DELTA-RHO	Relational vacuum response	D10-V0	differences and responses	nontrivial quotient invariant	physical relational datum	physical witness	SYM10-Q12
SYM10-ATLAS-LAMBDAEFF	Total gravitational coefficient	D10-V0	semiclassical field equation	nontrivial matched coefficient	boundary or empirical status	matched coefficient	SYM10-Q13
SYM10-ATLAS-STANDING	Standing or quotient correspondence	D10-V0	scalar prediction class	equivalence at restricted target	proved only in V0	realized bridge	SYM10-A04

D.6 Parameter role and mobility projection

Quantity	Role	Mobility	Witness	Blocked inference
$g(\mu)$ μ	transport-defined presentation or transport coordinate	transport running presentation change	RG trajectory plus observables Callan-Symanzik reconstruction	not free reassignment not a mass anchor
raw beta vector	coordinate transport vector	coordinate covariance plus possible vertical flavor/redundant motion	extended pushforward law	not a physical fixed-point criterion by itself
quotient or improved beta vector	physical transport structure	quotient stability	induced flow on nonredundant theory space	vanishing does not require every raw beta component to vanish
critical exponent matching scale	quotient witness effective or presentation coordinate	quotient stability effective drift	nonredundant stability spectrum overlap and matching certificate	not a raw matrix entry not automatically a threshold anchor
Λ_{RG}	transport-defined relation	quotient stability	RG-invariant trajectory relation	not numerically predicted without input
ρ_R in V_0	presentation-dependent offset	presentation change	finite-renormalization quotient	not a standalone prediction
Λ_{eff}	matched total coefficient	anchor non-runability at coupled target	semiclassical field equation	split not separately identifiable

D.7 Equation registry

ID	Relation	Normalized form	Theorem
E01	Callan-Symanzik equation	$(\mu d_\mu + \beta^i d_i + \gamma) G_R = 0$	Q03
E02	Scale-independent beta pushforward	$\beta_{prime} = J \beta$ for $g_{prime}=f(g)$ with no explicit μ dependence	Q04
E03	Operator coefficient pairing	$C_{prime} = C M^{-1}$	Q04
E04	Anomalous-dimension transport	$\gamma_{prime} = M \gamma M^{-1} - (\beta \cdot d M) M^{-1}$	Q04
E05	Trace identity	$T^{\mu}_{\mu} = \text{nonredundant beta contribution} + \text{mass} + \text{virial} + \text{improvement} + \text{EOM/contact} + \text{background}$	Q02
E06	Physical fixed-locus condition	$\overline{\beta}_\star([g_\star])=0$ on the nonredundant quotient	Q06
E07	Quotient stability matrix	$B^\alpha_\beta = d \overline{\beta}^\alpha / d \bar{g}^\beta$ at $[g_\star]$	Q07
E08	EFT matching	$A_{UV} = A_{IR} + O(Q/\Lambda)^{N+1}$	Q08
E09	Dimensional transmutation	$\Lambda_{RG} = \mu \exp(-\int dg/\beta)$	Q09
E10	Finite vacuum shift	$\rho_R \rightarrow \rho_R + \delta \rho$	Q12
E11	Semiclassical invariant	$\Lambda_{eff} = \Lambda_R + 8 \pi G_R \rho_R$	Q13
E12	Standing correspondence	Standing(X) iff X descends to V_0/R_{fin}	A04
E13	Explicit-scale scheme pushforward	$\beta_{prime}^A = d_B F^A \beta^B + \mu \partial_\mu F^A$ for $x_{prime}=F(x, \mu)$	Q04
E14	Quotient or improved beta	$\overline{\beta} = \pi_\star \beta$; locally $B^I = \beta^I - (Sg)^I$ when a flavor generator S represents the vertical flow	Q02;Q06;Q07

D.8 Bounded rival ledger

ID	Rival proposal	Disposition	Result	Reason
R10-01	Identify passive unit change with physical dilatation	incompatible	F01	acted-on object and quotient differ
R10-02	Transform a coupling but retain the old beta vector	incompatible	Q04	fails vector-field pushforward
R10-03	Rotate operators but not Wilson coefficients	incompatible	Q04	breaks the interaction pairing

ID	Rival proposal	Disposition	Result	Reason
R10-04	Declare fixed point from coordinate zero in one singular scheme	incomplete	Q06	requires quotient-stable regular chart
R10-05	Use equal exponents alone to define complete universality	incomplete	Q05	operator and symmetry data omitted
R10-06	Drop heavy fields without matching	incompatible	Q08	no continuation certificate
R10-07	Treat matching scale as measured threshold	incompatible	Q11;A03	coordinate is not an anchor without witness
R10-08	Infer numerical Lambda_RG from RG invariance	incompatible	Q09;Q11	boundary datum missing
R10-09	Promote IR symmetry to UV primitive	target error	Q10;A02	emergence ancestry omitted
R10-10	Use cutoff zero-point sum as direct curvature prediction	incompatible	Q12;Q13	absolute offset fails quotient and split is nonidentifiable
R10-11	Claim Casimir force measures absolute rho_R	incompatible	Q12	force is relational
R10-12	Extend standing or descent equivalence to all observables	target update	A04	theorem is restricted to V0
R10-13	Deny physical standing to a self-contained IR symmetry because no UV completion or matching map is specified	incompatible	Q10;A02	IR physicality and UV emergence provenance are distinct burdens

E Exact nonclaims and theorem boundary

ID	Blocked inference	Governing locus
NC10-01	No derivation of Standard Model beta functions from AASC	MAIN
NC10-02	No numerical prediction of observed couplings, masses, or thresholds	Q03;Q09;Q11
NC10-03	No claim that the observed Standard Model is scale or conformally invariant	Q01;Q06
NC10-04	No unrestricted implication from scale invariance to conformal invariance	Q06
NC10-05	No numerical determination of Lambda_RG without boundary, matching, or observable input	Q09
NC10-06	No hierarchy-problem solution	Q11
NC10-07	No claim that every vacuum-energy-like quantity is unphysical	Q12;Q13
NC10-08	No solution of cosmological-constant naturalness or radiative stability	Q12;Q13
NC10-09	No extension of the V0 standing correspondence without proof	A04
NC10-10	No classification of every fixed point or emergent symmetry	Q05;Q06;Q10
NC10-11	No treatment of RG flow as an ordinary internal symmetry on one fixed Hilbert space	F01;Q03
NC10-12	No candidate-completeness or exhaustion theorem	A05;MAIN
NC10-13	No inference from standalone IR symmetry physicality to emergence from a specified UV theory without a matching and ancestry certificate	Q10;A02

E.1 Boundary interpretation

The nonclaim registry is not ornamental. It fixes the strongest theorem actually proved. In particular:

- the quotient-transport theorem is conditional on overlap and matching data;
- the scale-to-conformal statement is premise-indexed;
- dimensional transmutation supplies an invariant relation, not an absolute numerical prediction;
- the vacuum correspondence is restricted to V_0 ;
- the artifact package does not establish semantic completeness.

A neighboring target can be scientifically important without being a counterexample to the theorem stated here.

F Reproducibility and bounded certification

F.1 Release contents

The project archive contains:

- the complete LaTeX manuscript and bibliography;
- stable target, theorem, import, source, atlas, rival, nonclaim, and hostile-audit registries;
- encoded illustrative algebra for scheme transport, fixed-point examples, transmutation, and the vacuum shift;
- generated publication tables;
- bounded validation and hostile-audit scripts;
- a release manifest, source map, notes, and build instructions.

Certificate

The bounded release validator passed **776/776** checks. The checks cover target and theorem identity, source wiring, selected encoded algebra, publication-table synchronization, release language, and artifact structure. They do not establish unencoded QFT, fixed-point existence, conformal enhancement, or semantic exhaustion.

Certificate

The bounded adversarial suite passed **23/23** declared corruption tests. The suite checks encoded covariance, target scope, no-promotion, matching, vacuum compensation, and artifact-authority boundaries. It does not reconstruct or formally prove the underlying QFT.

F.2 Certification doctrine

The scripts may verify file presence, stable IDs, source and citation wiring, table synchronization, selected algebraic identities, scope locks, and declared hostile corruptions. They do not infer a beta function, decide whether an interacting theory is conformal, prove decoupling, construct every universality class, or establish the curved-spacetime stress-tensor theorem. They are audit evidence, not theorem authority.

F.3 Build

A standard release build runs

```
make release
```

which generates tables, executes bounded validation, executes the hostile suite, compiles the PDF with `latexmk`, and copies the release-named manuscript. The release manifest is generated only after the final audit outputs and PDF are fixed.

G Bounded hostile-audit registry

ID	Mutation	Required detection	Locus
M10-01	Change coupling without beta pushforward	reject	Q04
M10-02	Rotate operators without coefficient transport	reject	Q04
M10-03	Omit connection term in gamma transport	reject	Q04
M10-04	Call an arbitrary subtraction point a mass anchor	reject	Q11
M10-05	Identify an EFT threshold with silent same-envelope continuation	reject	Q08
M10-06	Delete matching uncertainty or truncation order	reject	Q08

ID	Mutation	Required detection	Locus
M10-07	Promote an IR emergent symmetry to UV exactness	reject	Q10
M10-08	Infer conformal invariance without virial and stress premises	reject	Q06
M10-09	Keep a redundant direction in the critical spectrum	reject	Q07
M10-10	Treat equal exponents as a complete universality certificate	reject	Q05
M10-11	Infer numerical Lambda_RG from trajectory invariance	reject	Q09;Q11
M10-12	Change rho_R without compensating Lambda_R	reject	Q13
M10-13	Call a Casimir response an absolute rho_R measurement	reject	Q12
M10-14	Extend the V0 theorem to the full stress tensor	reject	Q12;A04
M10-15	Use naturalness as an independent scale selector	reject	Q11
M10-16	Treat validator success as proof of a QFT theorem	reject	certification boundary
M10-17	Delete a theorem registry ID	reject	release identity
M10-18	Corrupt dimensional-transmutation sign	reject	Q09
M10-19	Corrupt the semiclassical compensation sign	reject	Q13
M10-20	Replace A04 equivalence by an unrestricted QFT-wide claim	reject	A04
M10-21	Require every raw beta component to vanish at a quotient fixed class	reject	Q02;Q06;Q07
M10-22	Require a UV matching map for the physicality of a standalone IR symmetry	reject	Q10;A02
M10-23	Apply the scale-independent beta pushforward formula to an explicitly mu-dependent scheme map	reject	Q04

G.1 What a hostile scientific objection must do

A substantive objection must break a local derivation, invalidate an imported theorem or its specialization, identify a target mismatch, show that a claimed presentation changes physical witnesses, or provide a faithful same-target counterexample. The following do not suffice:

- demanding a full formalization of interacting QFT;
- objecting that a theorem is conditional when its premises are explicit;
- presenting a different EFT, state, threshold, or vacuum sector as same-target;
- requiring SYM-11’s whole-atlas exhaustion inside SYM-10;
- treating an artifact defect as a scientific refutation when the artifact is not proof authority.

G.2 Verdict rule

A scientific or dependency break can weaken or defeat the endpoint. A bounded artifact inconsistency normally requires an artifact repair. The two are linked only when the manuscript explicitly uses the artifact as a load-bearing premise, which this release does not.

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